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2224/73253; H01L 2224/73259; H01L
2224/73267; H01L 2224/13022; H01L
2224/29299; H01L 2224/83191; H01L
2224/2929; H01L 2224/16238; H01L
2224/293; H01L 2224/73265; H01L
2224/92224;

[illegible]

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H01L 23/522 (2006.01)
H01L 23/528 (2006.01)

(52) **U.S. Cl.**

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(2013.01); *H01L 23/528* (2013.01); *H01L*
24/94 (2013.01)

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H01L 2224/2919; H01L 2924/15311;
H01L 2924/19105; H01L 2224/24105;
H01L 2224/97; H01L 2224/16148; H01L
2924/181; H01L 24/14; H01L 23/28;
H01L 23/4952; H01L 23/5226; H01L
23/528; H01L 24/94; H01L 2225/1058;
H01L 24/32; H01L 24/48; H01L
2224/12105; H01L 23/49811; H01L
23/49894; H01L 25/105; H01L 23/3128;
H01L 2221/68359; H01L 2224/32145

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

9,048,222 B2	6/2015	Hung et al.	
9,048,233 B2	6/2015	Wu et al.	
9,064,879 B2	6/2015	Hung et al.	
9,111,949 B2	8/2015	Yu et al.	
9,263,511 B2	2/2016	Yu et al.	
9,281,254 B2	3/2016	Yu et al.	
9,299,649 B2	3/2016	Chiu et al.	
9,368,460 B2	6/2016	Yu et al.	
9,372,206 B2	6/2016	Wu et al.	
9,425,126 B2	8/2016	Kuo et al.	
9,443,783 B2	9/2016	Lin et al.	
9,461,018 B1	10/2016	Tsai et al.	
9,496,189 B2	11/2016	Yu et al.	
9,666,502 B2	5/2017	Chen et al.	
9,735,131 B2	8/2017	Su et al.	
2017/0033009 A1 *	2/2017	Scanlan	 H01L 23/48
2017/0345764 A1 *	11/2017	Chang	 H01L 23/5383
2018/0047674 A1 *	2/2018	Tsai	 H01L 24/19
2018/0076184 A1 *	3/2018	Chen	 H01L 24/73
2018/0269188 A1 *	9/2018	Yu	 H01L 23/3128
2020/0098621 A1 *	3/2020	Bharath	 H01L 25/18

* cited by examiner

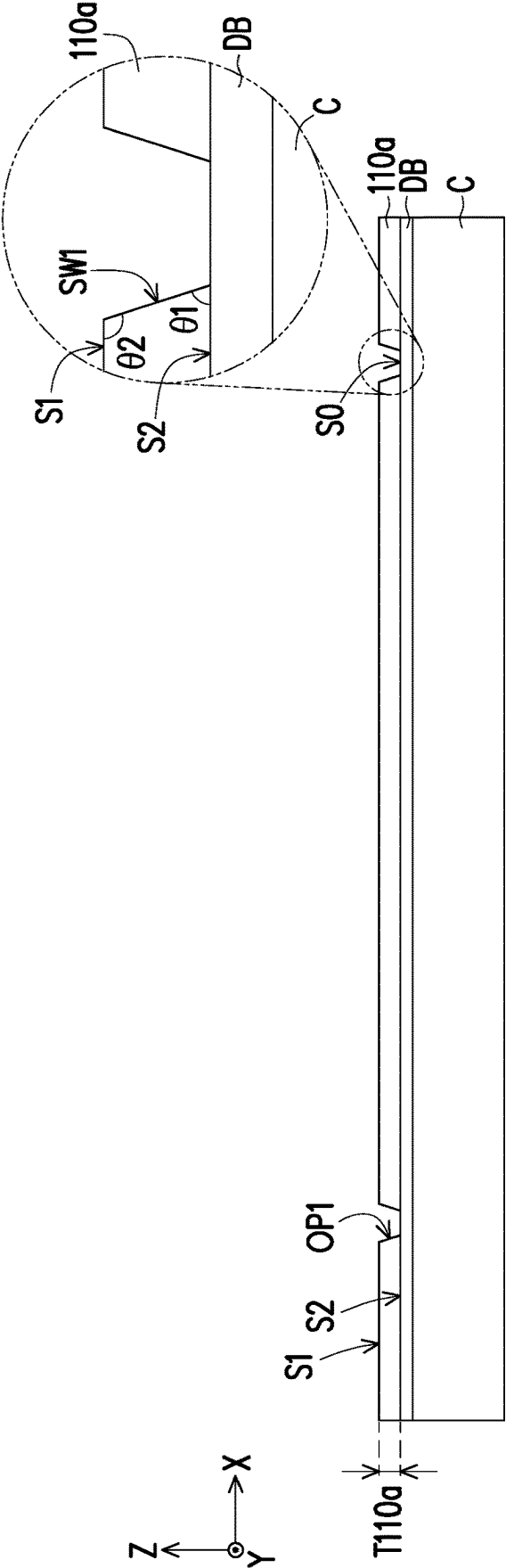


FIG. 1

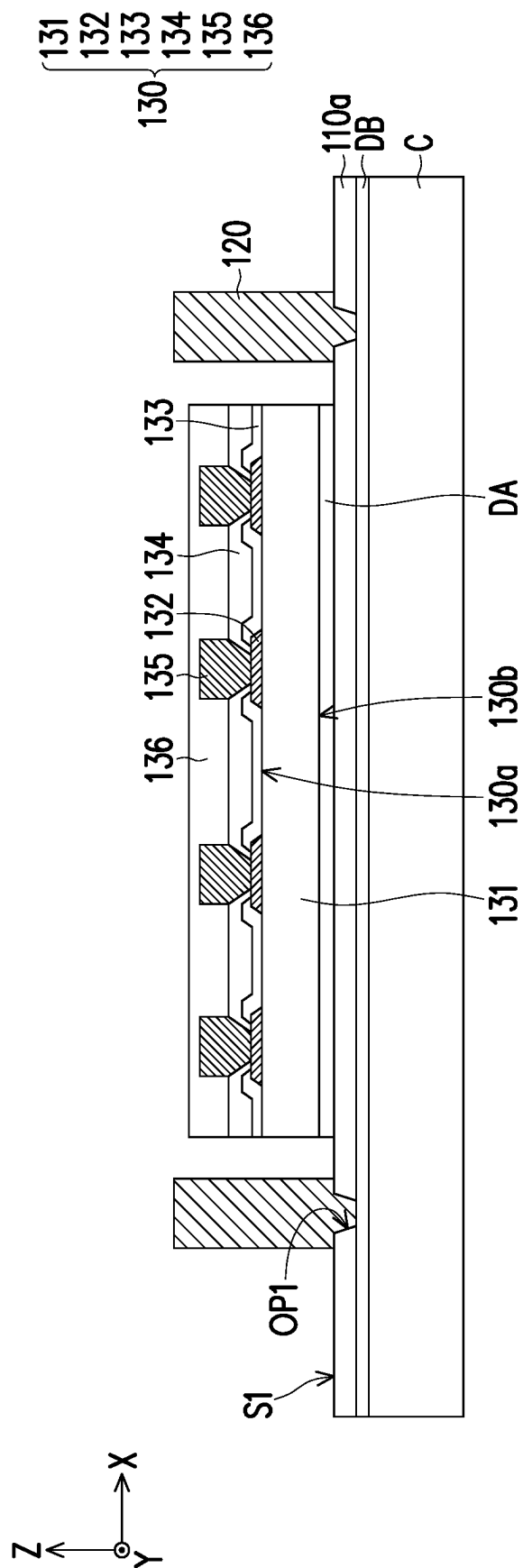


FIG. 2

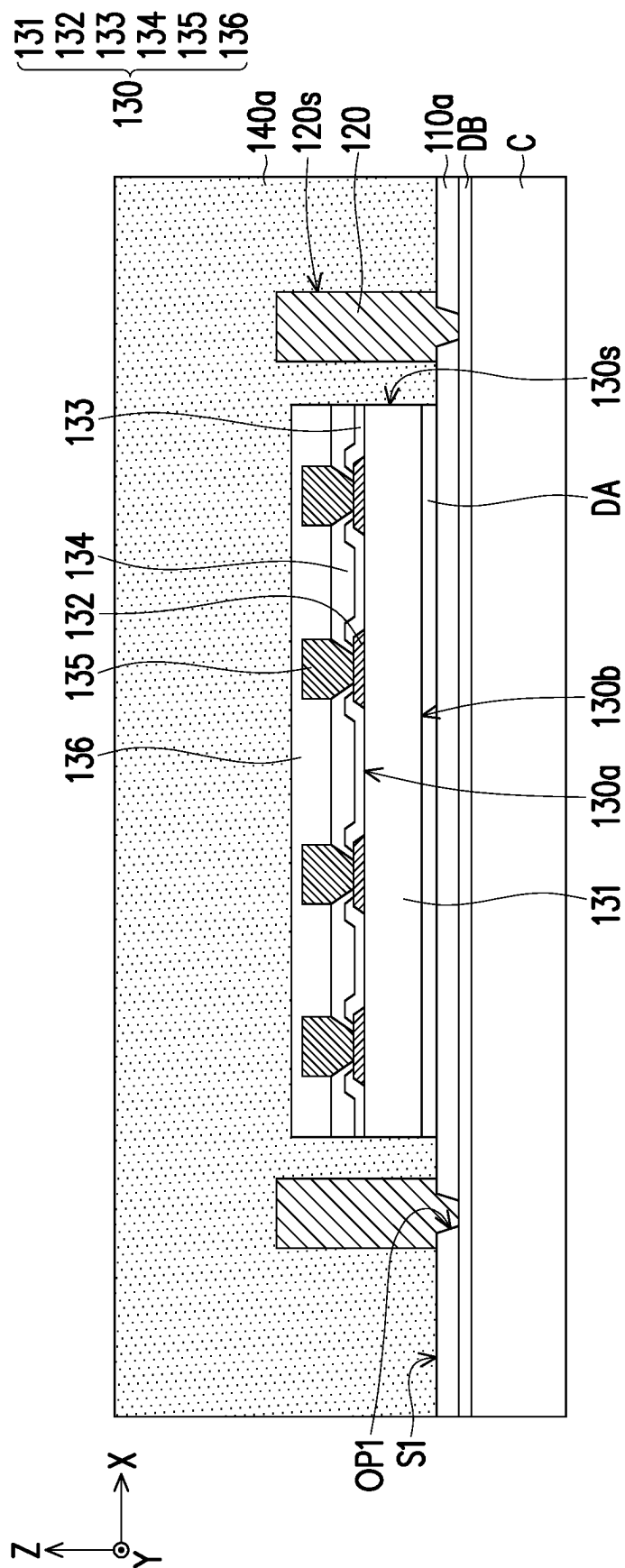


FIG. 3

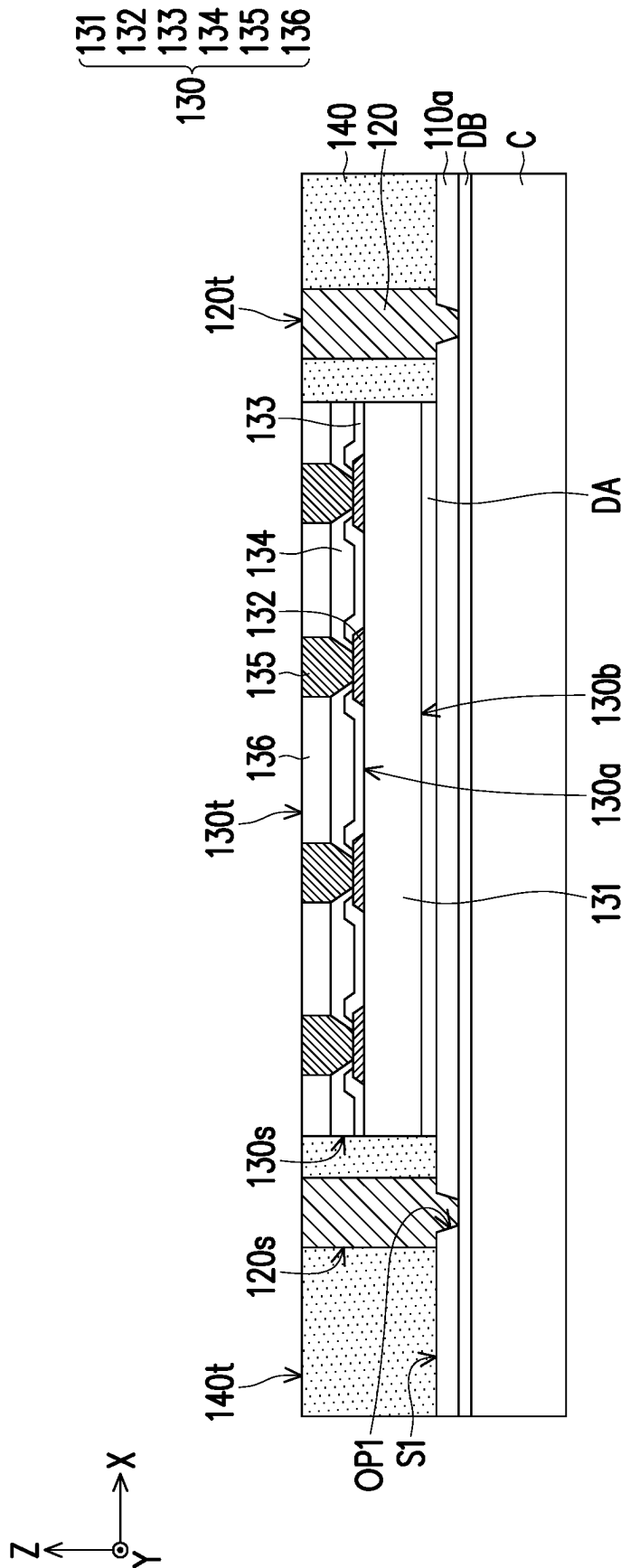


FIG. 4

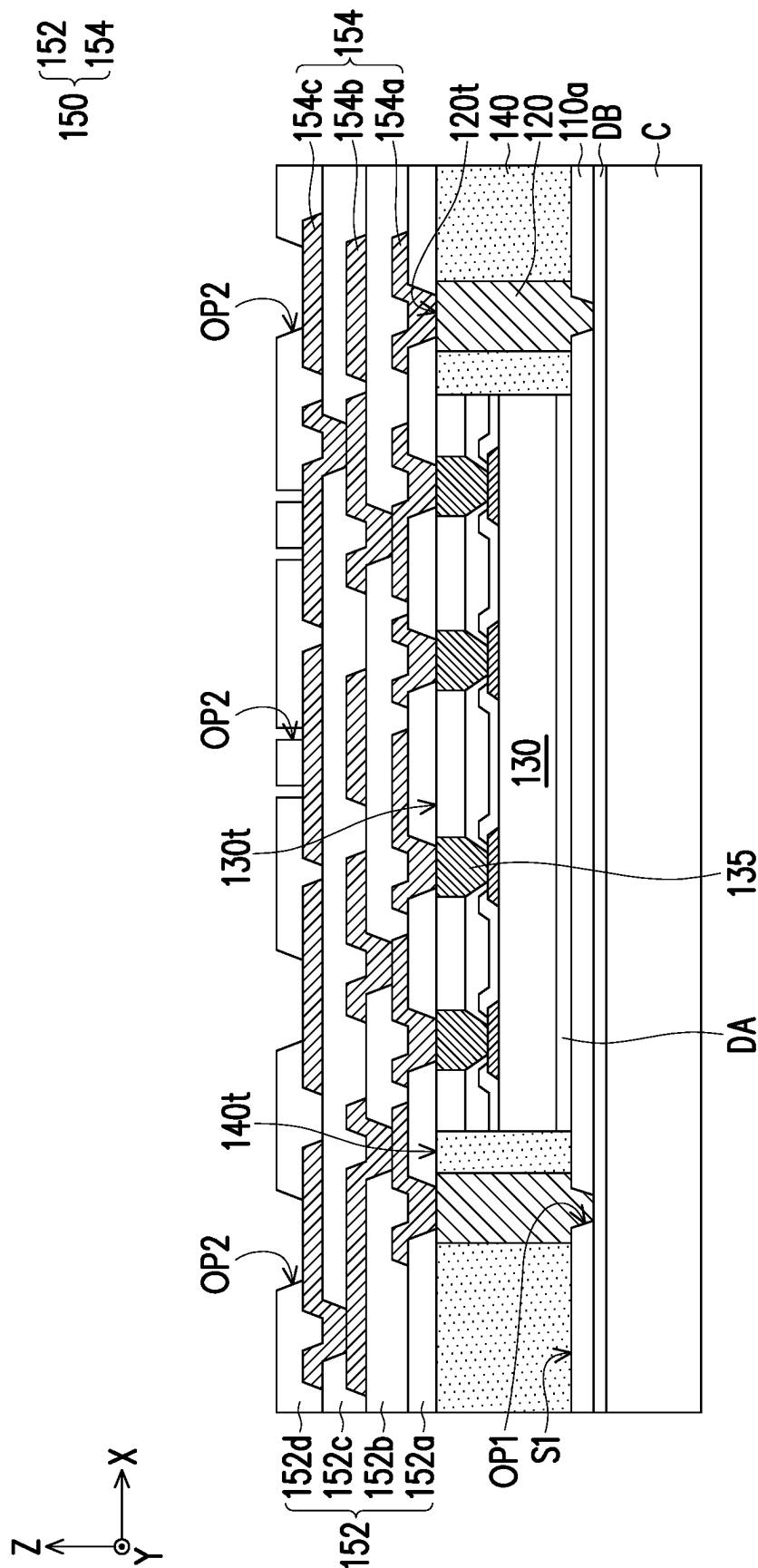


FIG. 5

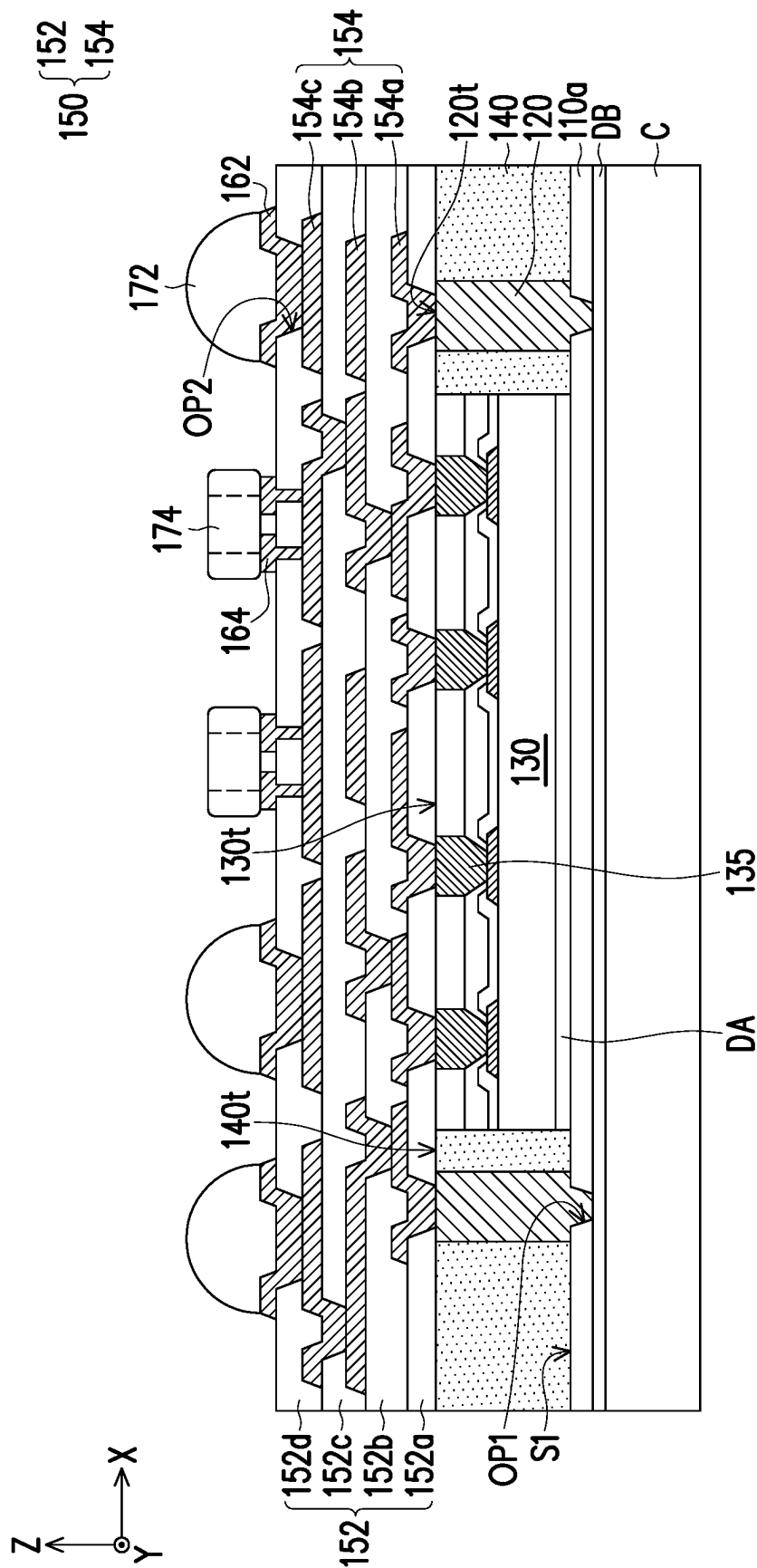


FIG. 6

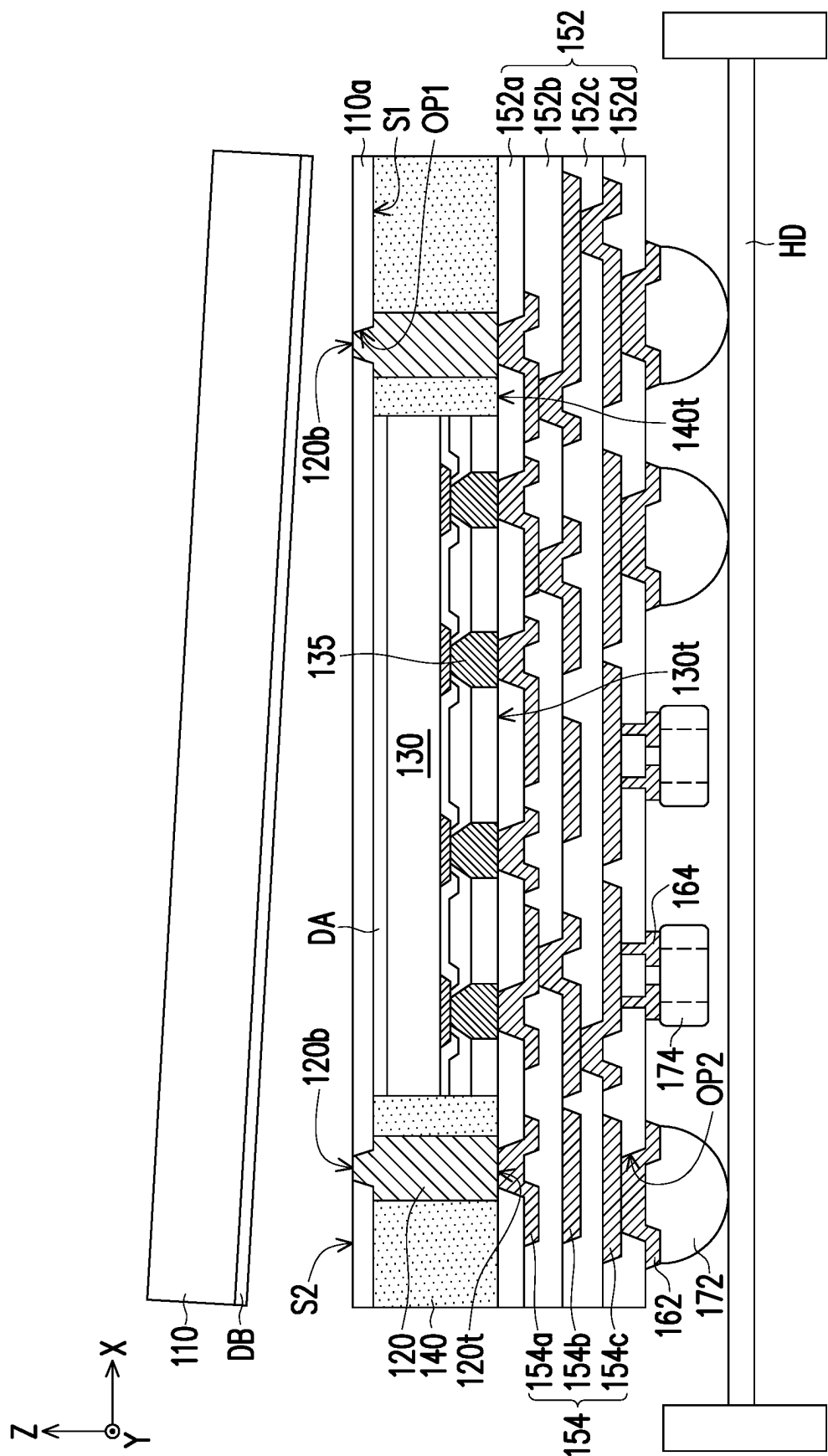


FIG. 7

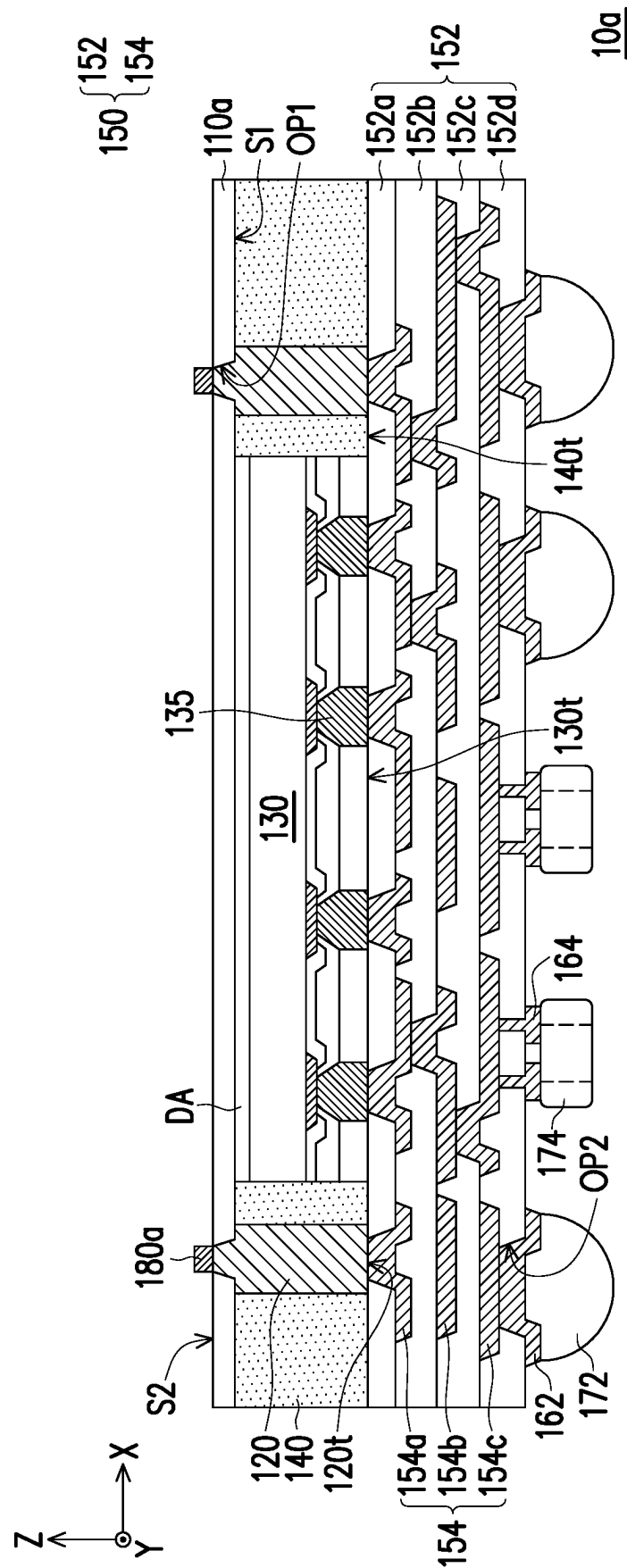


FIG. 8

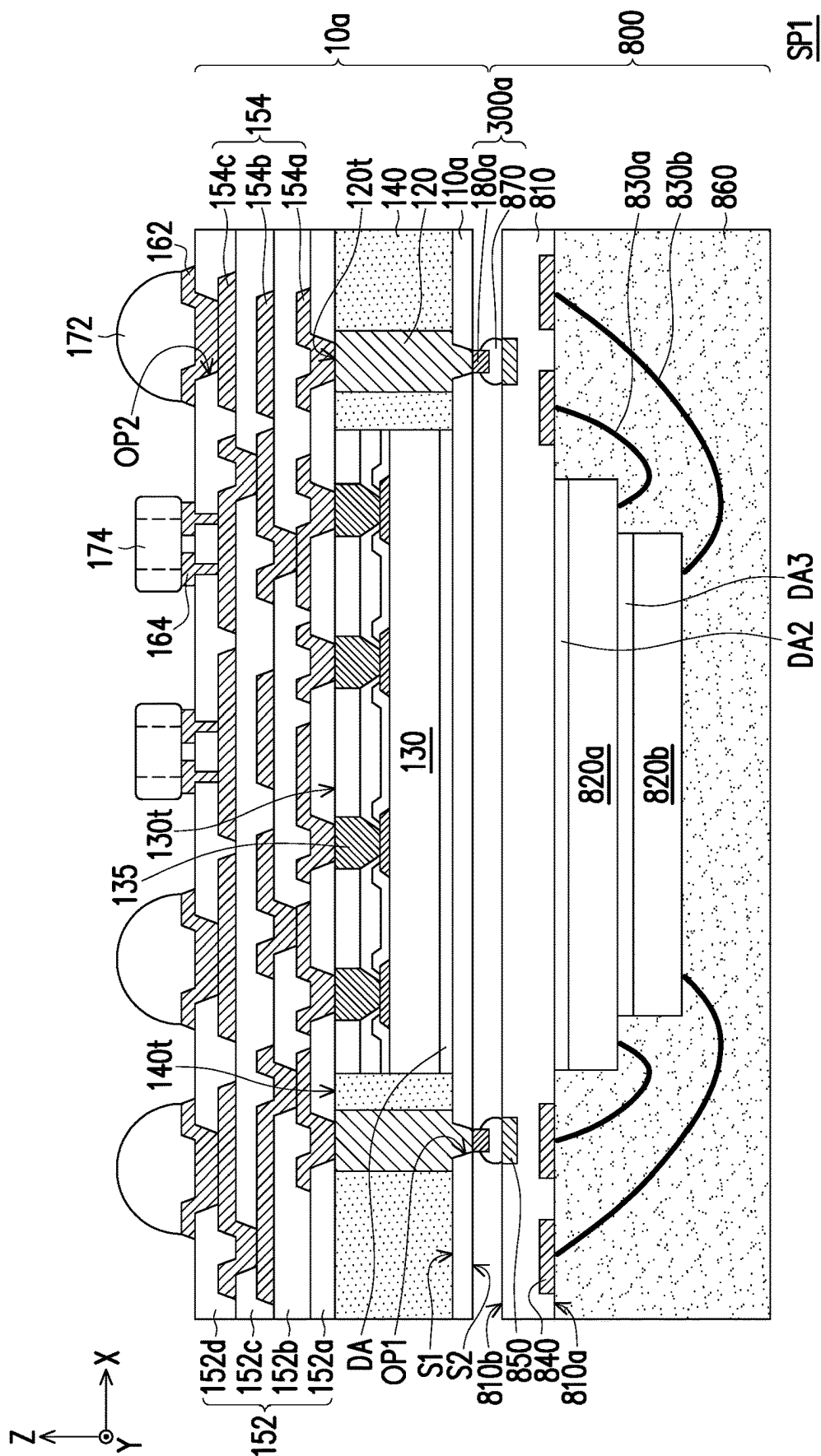
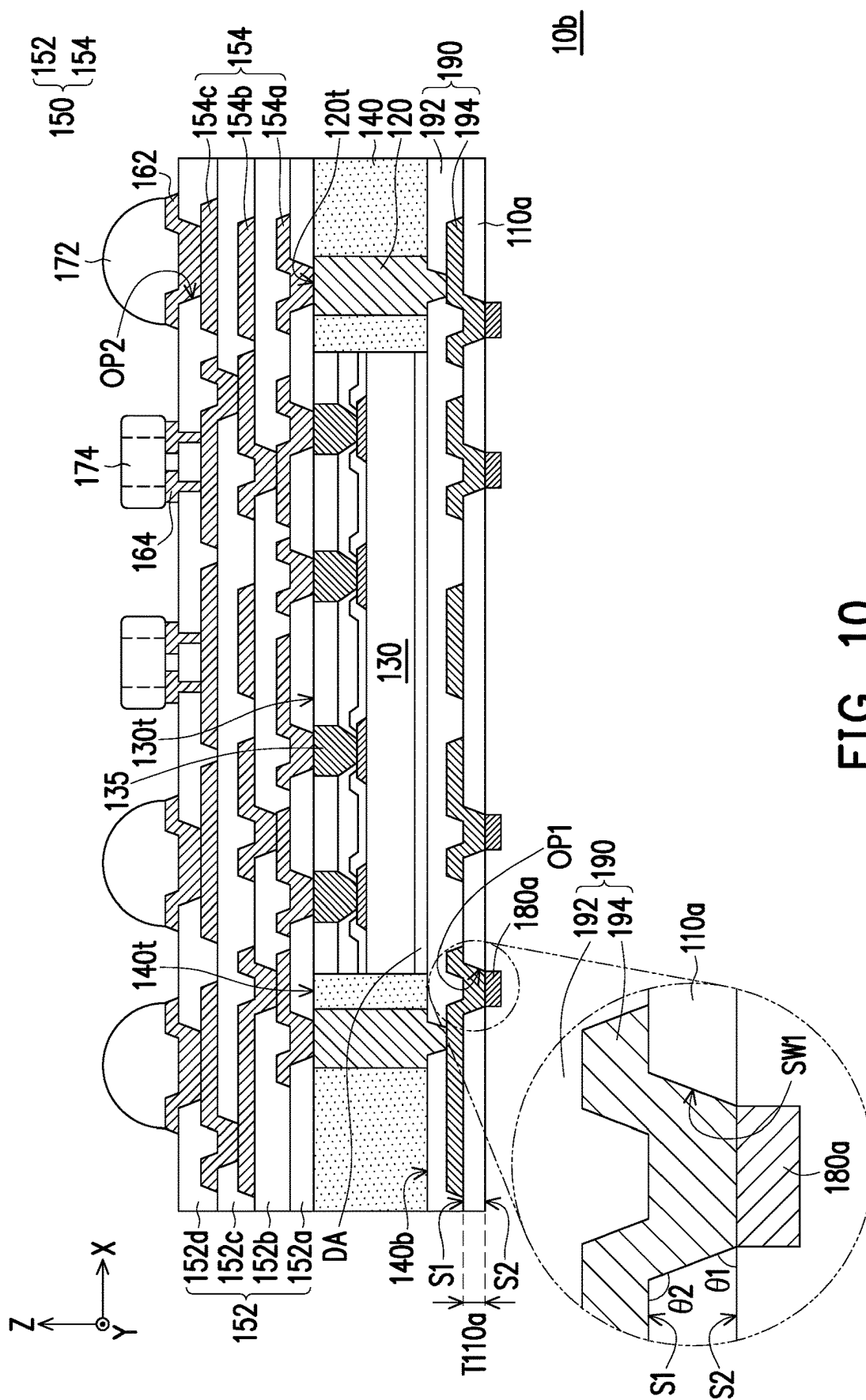


FIG. 9



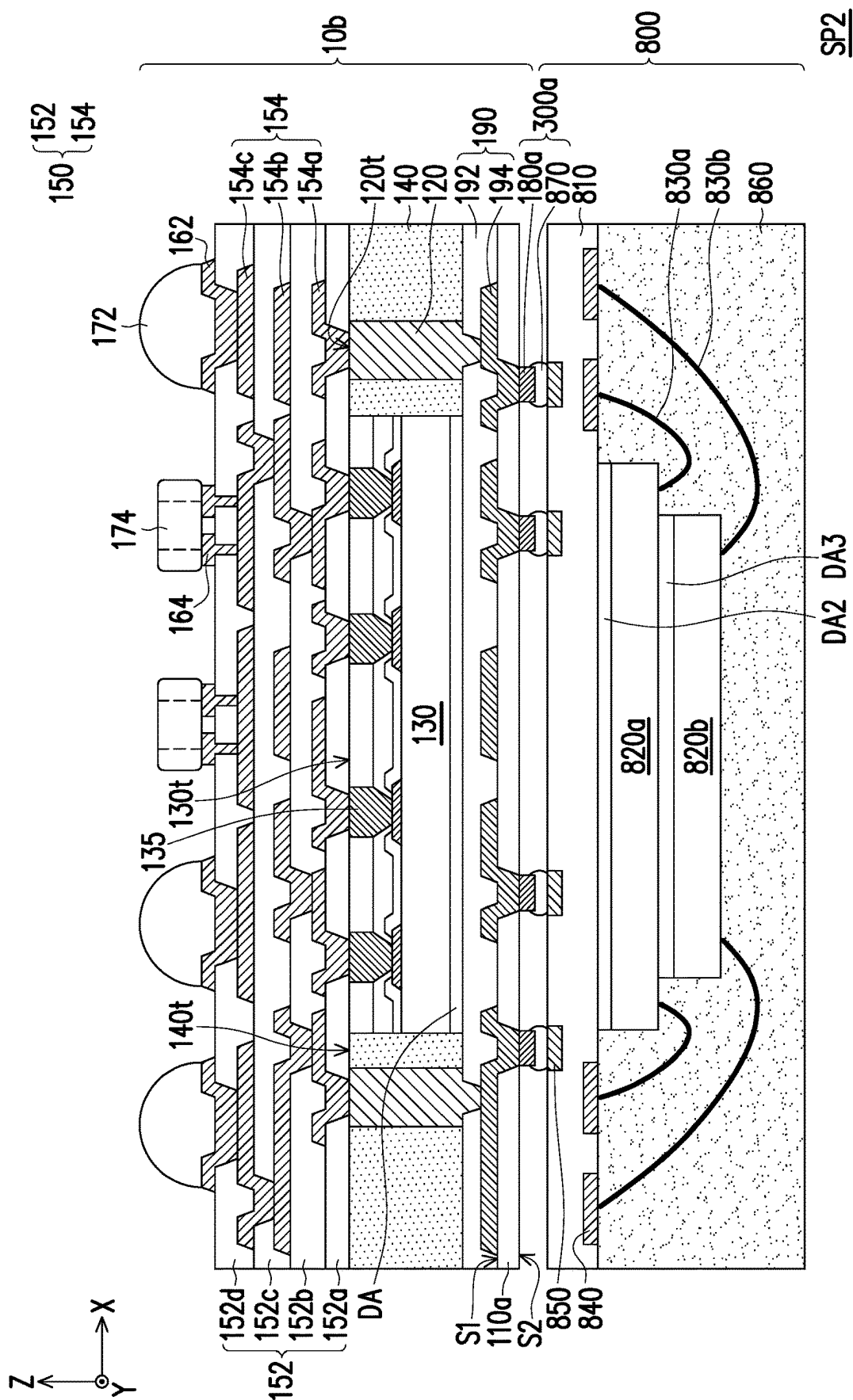


FIG. 11

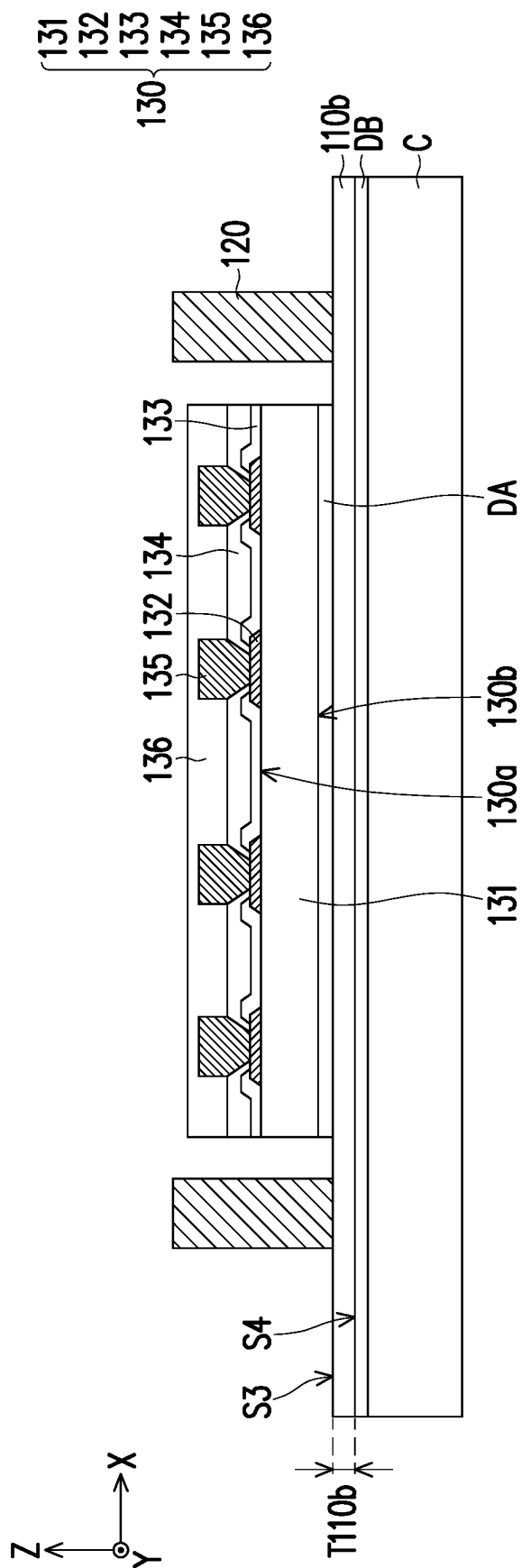


FIG. 12

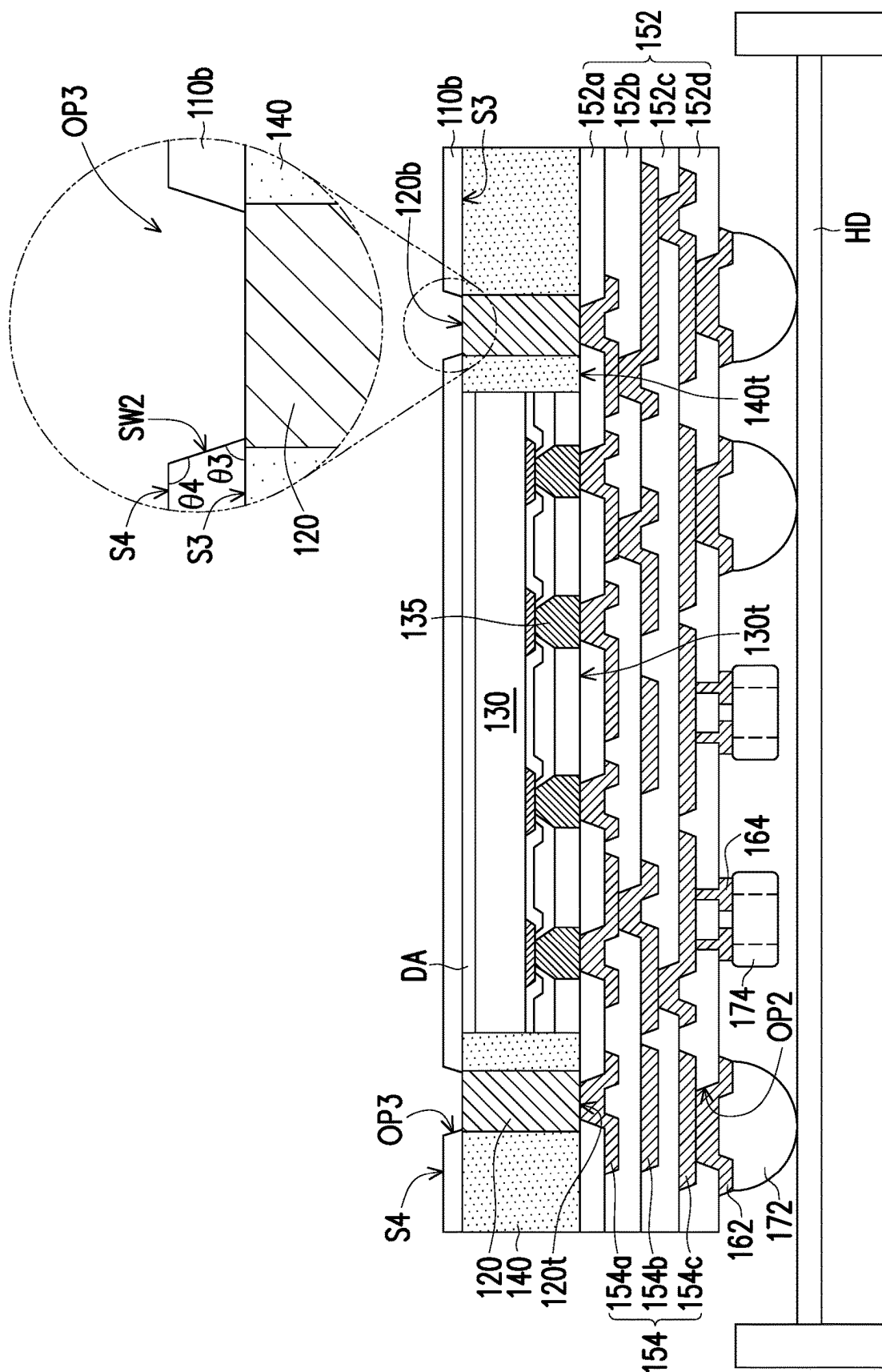


FIG. 13

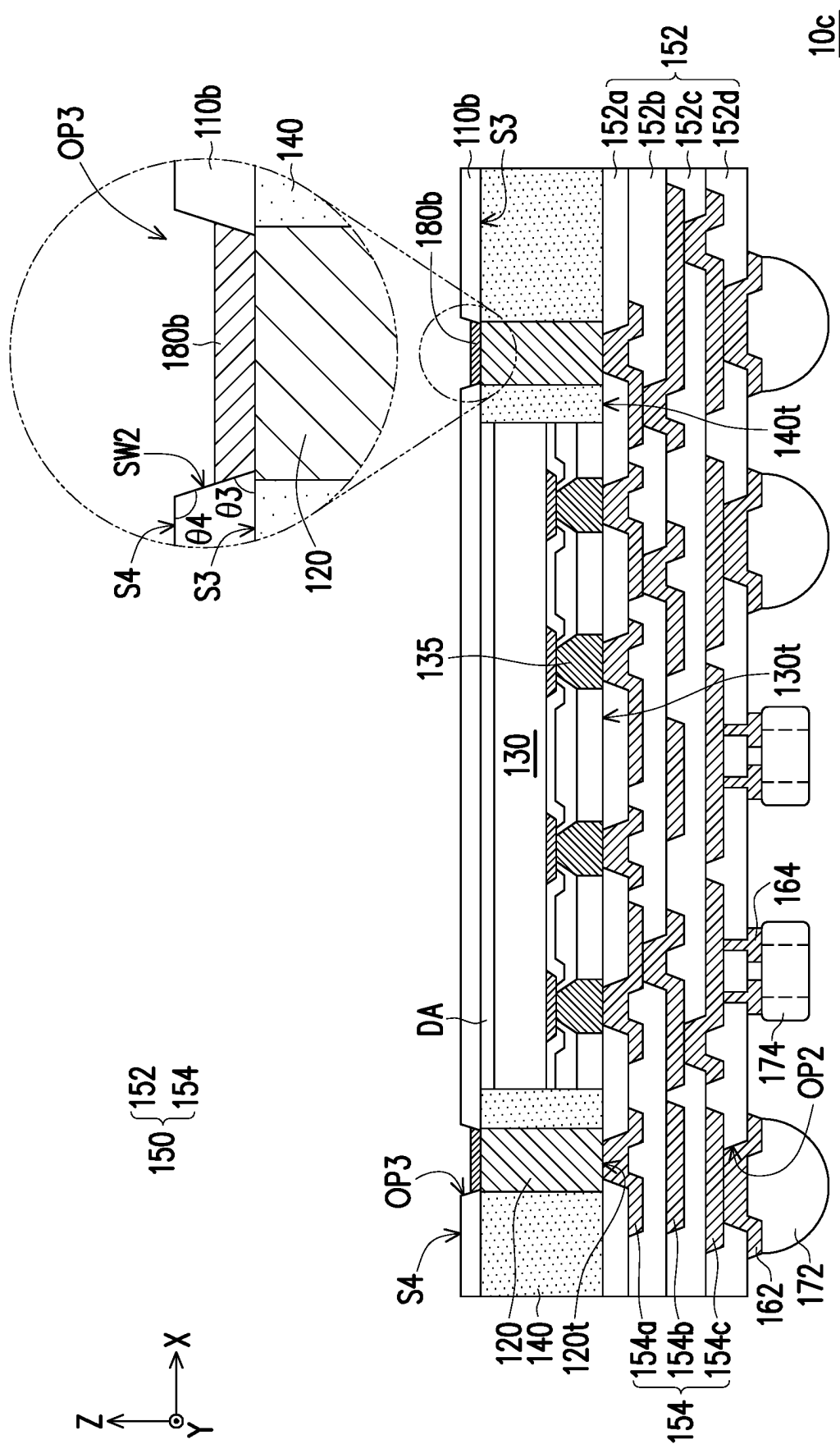


FIG. 14

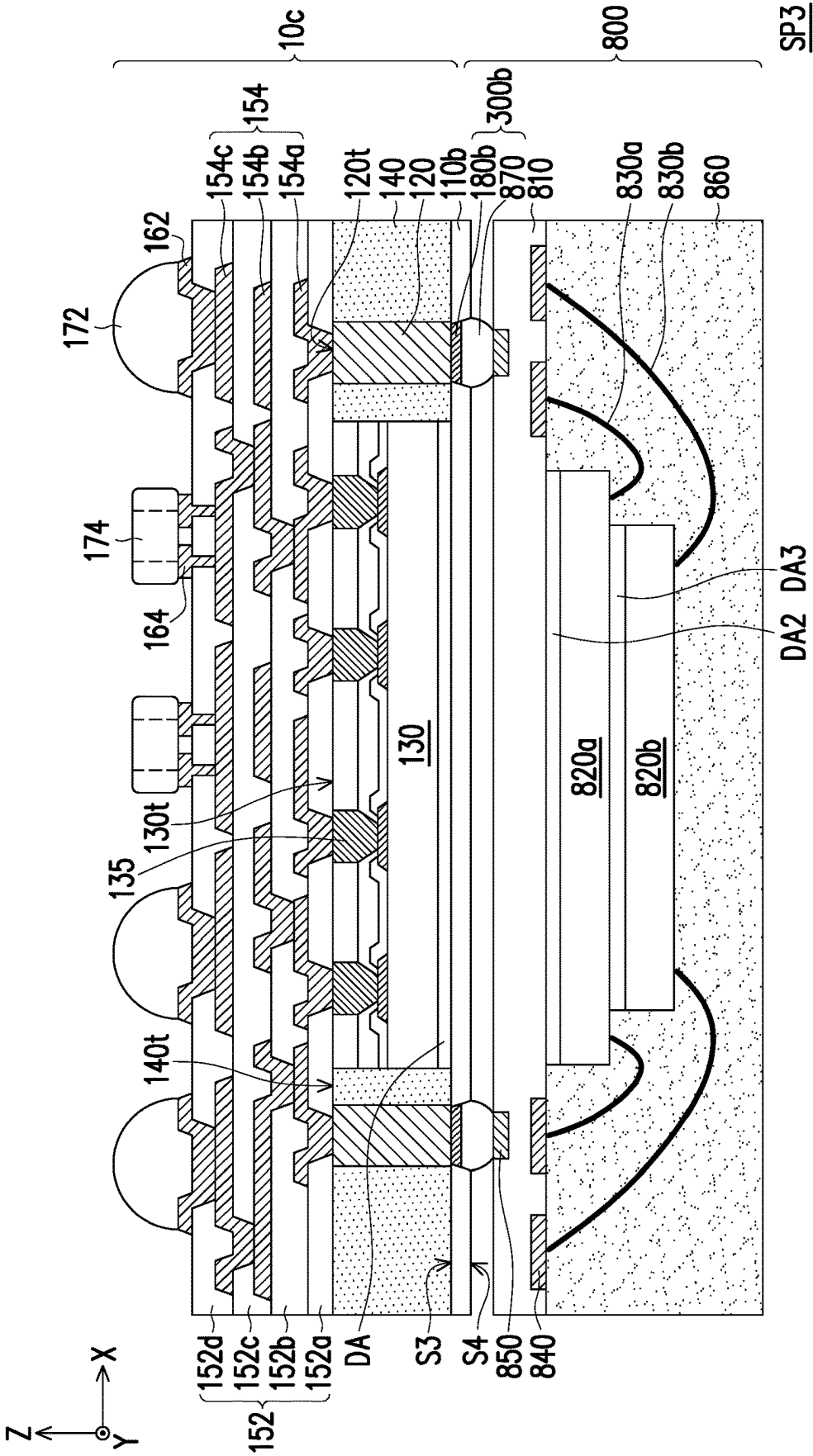


FIG. 15

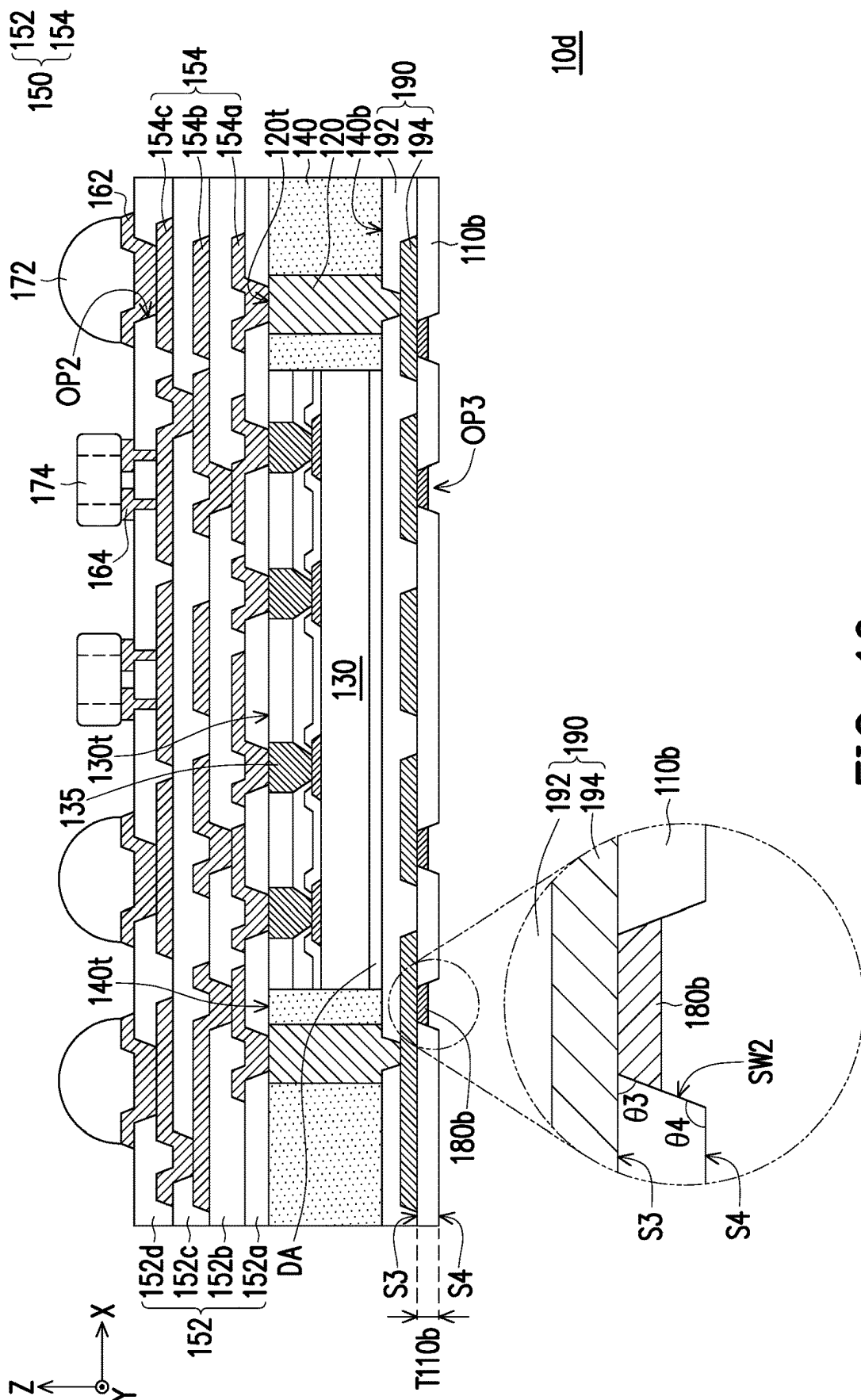


FIG. 16

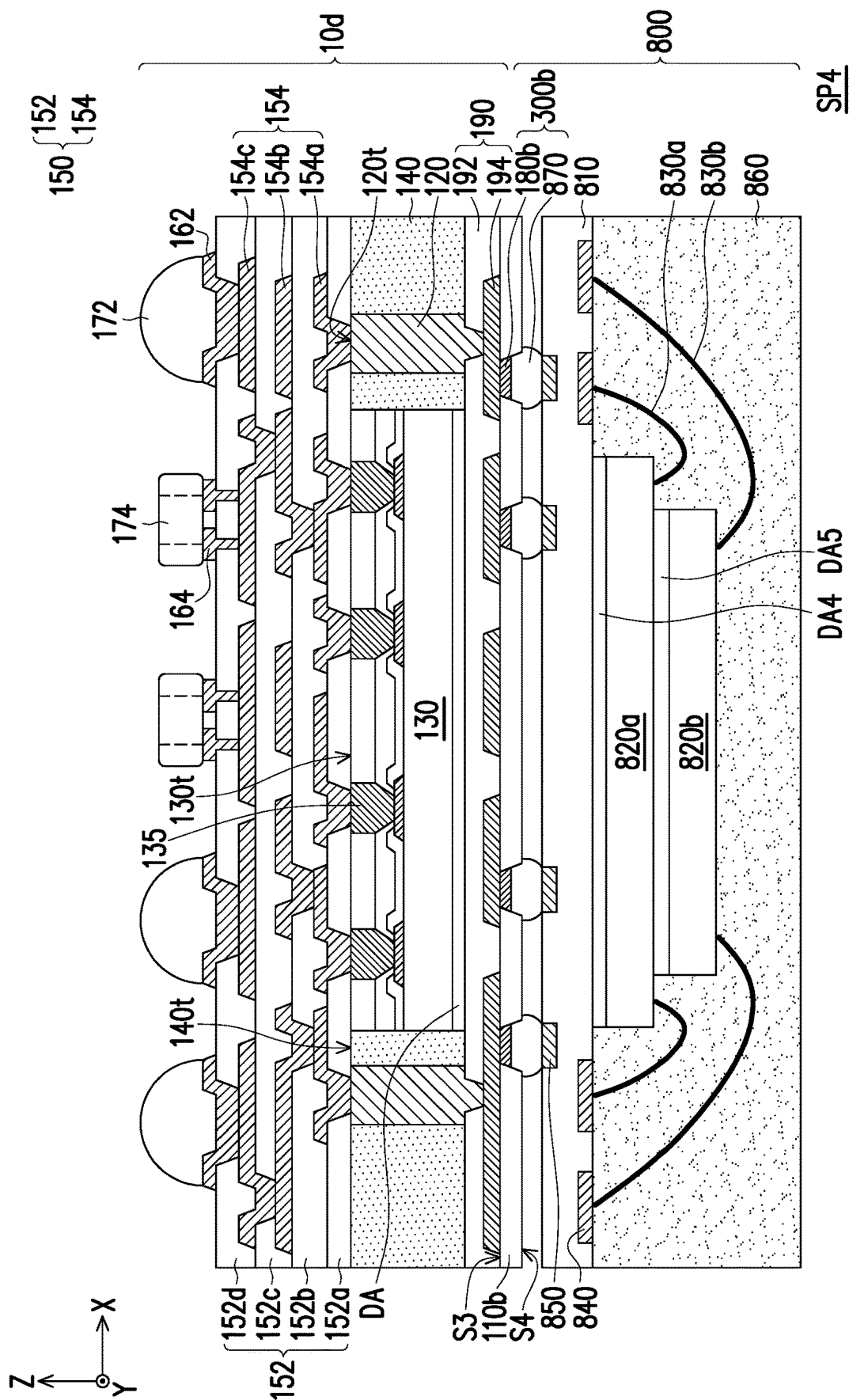


FIG. 17

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PACKAGE STRUCTURE AND METHOD OF MANUFACTURING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 16/718,213, filed on Dec. 18, 2019. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Semiconductor devices and integrated circuits are typically manufactured on a single semiconductor wafer. The dies of the wafer may be processed and packaged with other semiconductor devices or dies at the wafer level, and various technologies have been developed for the wafer level packaging.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 to FIG. 8 are schematic cross sectional views of various stages in a manufacturing method of a package structure in accordance with some embodiments of the disclosure.

FIG. 9 is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure.

FIG. 10 is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure.

FIG. 11 is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure.

FIG. 12 to FIG. 14 are schematic cross sectional views of various stages in a manufacturing method of a package structure in accordance with some embodiments of the disclosure.

FIG. 15 is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure.

FIG. 16 is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure.

FIG. 17 is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components, values, operations, materials, arrangements, or the like, are described below to simplify the disclosure. These are, of course, merely examples and are not intended to be limiting. Other components, values, operations, materials, arrangements, or the like, are contemplated. For

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example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

In addition, terms, such as “first,” “second” and the like, may be used herein for ease of description to describe similar or different element(s) or feature(s) as illustrated in the figures, and may be used interchangeably depending on the order of the presence or the contexts of the description.

Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

FIG. 1 to FIG. 8 are schematic cross sectional views of various stages in a manufacturing method of a package structure in accordance with some embodiments of the disclosure. In embodiments, the manufacturing method is part of a wafer level packaging process. It is to be noted that the process steps described herein cover a portion of the manufacturing processes used to fabricate a package structure. The embodiments are intended to provide further explanations but are not used to limit the scope of the present disclosure. In some embodiments, as shown in FIG. 1 to FIG. 8, one (semiconductor) chip or die is shown to represent plural (semiconductor) chips or dies of the wafer, and one (semiconductor) package structure is shown to represent plural (semiconductor) package structures obtained following the (semiconductor) manufacturing method, however the disclosure is not limited thereto. In alternative embodiments, more than one (semiconductor) chips or dies are shown to represent plural (semiconductor) chips or dies of the wafer, and one or more than one (semiconductor) package structure are shown to represent plural (semiconductor) package structures obtained following the (semiconductor) manufacturing method.

Referring to FIG. 1, in some embodiments, a carrier C with a debond layer DB and a solder resist layer 110a coated thereon is provided. In some embodiments, the carrier C may be a glass carrier or any suitable carrier for carrying a semiconductor wafer or a reconstituted wafer for the manufacturing method of the semiconductor package.

In some embodiments, the debond layer DB is disposed on the carrier C, as shown in FIG. 1. The material of the debond layer DB may be any material suitable for bonding and debonding the carrier C from the above layer(s) (e.g. the solder resist layer **110a**) or any wafer(s) disposed thereon. In some embodiments, the debond layer DB may include a dielectric layer made of a dielectric material including any suitable polymer-based dielectric material (such as benzocyclobutene (BCB), polybenzoxazole (PBO)). In an alternative embodiment, the debond layer DB may include a dielectric material layer made of an epoxy-based thermal-release material, which loses its adhesive property when heated, such as a light-to-heat-conversion (LTHC) release coating film. In a further alternative embodiment, the debond layer DB may include a dielectric material layer made of an ultra-violet (UV) glue, which loses its adhesive property when exposed to UV lights. In certain embodiments, the debond layer DB may be dispensed as a liquid and cured, or may be a laminate film laminated onto the carrier C, or may be the like. The top surface of the debond layer DB, which is opposite to a bottom surface contacting the carrier C, may be levelled and may have a high degree of coplanarity. In certain embodiments, the debond layer DB is, for example, a LTHC layer with good chemical resistance, and such layer enables room temperature debonding from the carrier C by applying laser irradiation, however the disclosure is not limited thereto.

In some embodiments, the solder resist layer **110a** is disposed on the debond layer DB, and the debond layer DB is located between the carrier C and the solder resist layer **110a**. In some embodiments, a surface S1 (e.g. a top surface) of the solder resist layer **110a** may provide a high degree of coplanarity and flatness. Due to the high degree of coplanarity and flatness, the formation of the later-formed layer(s) and/or element(s) is beneficial. As shown in FIG. 1, along a direction Z (e.g. a stacking direction of the carrier C, the debond layer DB and the solder resist layer **110a**), a thickness **T110a** of the solder resist layer **110a** is approximately from 10 μm to 30 μm .

In some embodiments, the solder resist layer **110a** is a layer made of a solder resist material, where the solder resist material is composed of an epoxy-based resin and a filler. In the solder resist layer **110a**, a weight percentage ratio of the epoxy-based resin to the filler is approximately from 40:60 to 60:40, in some embodiments. The filler, for example, includes silica (SiO_2), barium sulfate (BaSO_4), or a combination thereof, where a particle diameter of the filler is approximately from 0.2 μm to 2 μm . In some embodiments, the solder resist layer **110a** is formed on the debond layer DB by lamination. In some embodiments, the solder resist layer **110a** has a coefficient of thermal expansion (CTE) approximately ranging from 18 ppm/K to 35 ppm/K, a Young's modulus (E) approximately ranging from 5 GPa to 10 GPa, and a glass transition temperature (T_g) approximately ranging from 150 degrees Celsius to 180 degrees Celsius. In certain embodiments, the solder resist layer **110a** is photosensitive (see package structures **10a** and **10b** respectively depicted in FIG. 6 and FIG. 10). However, the disclosure is not limited thereto; in alternative embodiments, the solder resist layer **110a** is non-photosensitive (see package structures **10c** and **10d** respectively depicted in FIG. 14 and FIG. 16). Due to the solder resist layer **110a** (e.g. low CTE value) in addition to its specific thickness range, better warpage control (e.g., warpage being less than or substantially equal to 80 μm at room temperature and being greater than or substantially equal to -80 μm) to the package structure **10a** is achieved.

As illustrated in FIG. 1, in the embodiment of which the solder resist layer **110a** is photosensitive, after the solder resist layer **110a** is laminated onto the debond layer DB with a surface S2 (e.g. a bottom surface), a plurality of openings OP1 are formed, by photolithography processes, in the solder resist layer **110a** to expose portions of the debond layer DB. For example, a surface S0 of the debond layer DB is partially exposed by the openings OP1 formed in the solder resist layer **110a**. In the solder resist layer **110a** shown in FIG. 1, an angle θ_1 between the surface S2 of the solder resist layer **110a** and a sidewall SW1 of each opening OP1 is approximately 60 degrees to 80 degrees, and an angle θ_2 (i.e. $\theta_2=180\text{ degrees}-\theta_1$) between the surface S1 of the solder resist layer **110a** and the sidewall SW1 of each opening OP1 is 100 degrees to 120 degrees. The surface S2 is opposite to the surface S1 along the direction Z, and the surface S2 is stacked on the surface S0 of the debond layer DB as shown in FIG. 1, for example. With the formation of the solder resist layer **110a** having the openings OP1, the manufacturing cost and process complexity are further reduced.

Only two openings OP1 are shown in FIG. 1 for illustrative purposes, and the disclosure is not limited thereto. The number of the openings OP1 may be more than two based on the demand and the design layout. Additionally, for example, on a X-Y plane (where a direction X is different from a direction Y, and the directions X and Y are different from the direction Z (e.g. the stacking direction)), dimensions (e.g. maximum widths) of the openings OP1 may be the same, however the disclosure is not limited thereto. In an alternative embodiment, according to the design layout and/or demand, the dimensions of the openings OP1 may be the different from each other or may be different in a manner of different groups. In one embodiment, on the X-Y plane, a cross-sectional shape of the openings OP1 individually may be round, elliptical, oval, tetragonal, octagonal or any suitable polygonal shape; the disclosure is not limited thereto.

Referring to FIG. 2, in some embodiments, at least one conductive pillar **120** and at least one semiconductor die **130** are formed on the solder resist layer **110a**. For illustrative purposes, the at least one conductive pillar **120** include a plurality of conductive pillars **120** (e.g. two conductive pillars **120**), and at least one semiconductor die **130** include one semiconductor die **130**, as presented in FIG. 2. However, the number of the conductive pillars **120** and the number of the semiconductor die **130** are not limited to what is depicted in the disclosure, and may be selected and designated based on the demand and design layout. For example, the number of the conductive pillars **120** may be more than two and the number of the semiconductor die **130** may be more than one, where the number of the conductive pillars **120** may be adjusted by changing the number of the openings OP1. In some embodiments, the conductive pillars **120** and the semiconductor die **130** are arranged side-by-side on the solder resist layer **110a**.

In some embodiments, the conductive pillars **120** are formed on the solder resist layer **110a** (e.g. the surface S1 of the solder resist layer **110a**). In some embodiments, the conductive pillars **120** may be through integrated fan-out (InFO) vias. In some embodiments, the conductive pillars **120** are arranged along but not on a cutting line (not shown) between two package structures (e.g. two of the package structures **10a**). As shown in FIG. 2, the conductive pillars **120** are formed on the solder resist layer **110a** and penetrate through the solder resist layer **110a** via the openings OP1, in

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some embodiment. Through the openings OP1, the conductive pillars 120 are in physical contact with the debond layer DB, for example.

In some embodiments, the conductive pillars 120 are formed by photolithography, plating, photoresist stripping processes or any other suitable method. For example, the plating process may include an electroplating plating, an electroless plating, or the like. For example, the conductive pillars 120 may be formed by forming a mask pattern (not shown) covering the solder resist layer 110a with openings exposing the surface S0 of the debond layer DB exposed by the openings OP1 formed in the solder resist layer 110a, forming a metallic material filling the openings formed in the mask pattern and the openings OP1 to form the conductive pillars 120 by electroplating or deposition and then removing the mask pattern. In one embodiment, the mask pattern may be removed by acceptable ashing process and/or photoresist stripping process, such as using an oxygen plasma or the like. In some embodiments, prior to the formation of the mask pattern, a seed layer (not shown) may be formed conformally over the solder resist layer 110a and extend into the openings OP1 to be located on the debond layer DB, where the metallic material filling the openings formed in the mask pattern and the openings OP1 formed in the solder resist layer 110a is used as a mask to remove portions of the seed layer not being covered thereto. The disclosure is not limited thereto. In some embodiments, the material of the conductive pillars 120 may include a metal material such as copper or copper alloys, or the like. Throughout the description, the term "copper" is intended to include substantially pure elemental copper, copper containing unavoidable impurities, and copper alloys containing minor amounts of elements such as tantalum, indium, tin, zinc, manganese, chromium, titanium, germanium, strontium, platinum, magnesium, aluminum or zirconium, etc.

However, the disclosure is not limited thereto. In alternative embodiments, the conductive pillars 120 may be pre-fabricated conductive pillars which may be disposed on the solder resist layer 110a by picking- and placing.

Continued on FIG. 2, in some embodiments, the semiconductor die 130 is disposed on the solder resist layer 110a and over the carrier C. For example, the semiconductor die 130 is picked-up and placed on the solder resist layer 110a, and is attached or adhered on the solder resist layer 110a through a connecting film DA. In some embodiments, the connecting film DA is located between the semiconductor die 130 and the solder resist layer 110a, where the connecting film DA physically contacts the backside surface 130b of the semiconductor die 130 and the solder resist layer 110a (e.g. the surface S1 of the solder resist layer 110a). Due to the connecting film DA, the semiconductor die 130 and the solder resist layer 110a are stably adhered to each other. In some embodiments, the connecting film DA may be, for example, a die attach film, a layer made of adhesives or epoxy resin, or the like.

In some embodiments, the semiconductor die 130 includes a substrate 131 having an active surface 130a and a backside surface 130b opposite to the active surface 130a (along the direction Z), a plurality of conductive pads 132 formed on the active surface 130a, a passivation layer 133 disposed on and partially exposing the conductive pads 132, a post-passivation layer 134 disposed on the passivation layer 133 and partially exposing the conductive pads 132, connecting vias 135 disposed on the conductive pads 132, and a protection layer 136 covering the post-passivation layer 134 and the connecting vias 135. In other words, the conductive pads 132 distributed on the active surface 130a

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of the substrate 131 are partially exposed by contact openings of the passivation layer 133 and contact openings of the post-passivation layer 134, so as to physically connect to the connecting vias 135.

For example, the substrate 131 is a semiconductor substrate. In some embodiments, the material of the substrate 131 may include a silicon substrate including active components (e.g., transistors and/or memories such as NMOS and/or PMOS devices, or the like) and/or passive components (e.g., resistors, capacitors, inductors or the like) formed therein. In an alternative embodiment, the substrate 131 may be a bulk silicon substrate, such as a bulk substrate of monocrystalline silicon, a doped silicon substrate, an undoped silicon substrate, or a SOI substrate, where the dopant of the doped silicon substrate may be an N-type dopant, a P-type dopant or a combination thereof. The disclosure is not limited thereto.

In some embodiments, the conductive pads 132 may be aluminum pads or other suitable metal pads. For example, the conductive pads 132 may be formed by electroplating or deposition, and then patterned using a photolithography and etching process.

In some embodiments, the connecting vias 135 may be copper pillars, copper alloy pillar or other suitable metal pillars. For example, the forming process of the connecting vias 135 may be substantially the same or similar to the formation of the conductive pillars 120. However, the disclosure is not limited thereto.

In some embodiments, the passivation layer 133, the post-passivation layer 134 and/or the protection layer 136 may be a PBO layer, a polyimide (PI) layer or other suitable polymers. In certain embodiments, the passivation layer 133, the post-passivation layer 134 and/or the protection layer 136 may be made of inorganic materials, such as silicon oxide, silicon nitride, silicon oxynitride, or any suitable dielectric material. In one embodiment, the materials of the passivation layer 133, the post-passivation layer 134 and/or the protection layer 136 may be the same. In an alternative embodiment, the materials of the passivation layer 133, the post-passivation layer 134 and/or the protection layer 136 may be different from one another, the disclosure is not limited thereto.

In some embodiments, the semiconductor die 130 described herein may be referred to as a chip or an integrated circuit (IC). For example, in an alternative embodiment, the semiconductor die 130 includes a digital chip, an analog chip, or a mixed signal chip, such as an application-specific integrated circuit ("ASIC") chip, a sensor chip, a wireless and radio frequency (RF) chip, a memory chip, a logic chip, a voltage regulator chip, or a combination thereof. In an alternative embodiment, the semiconductor die 130 may be referred to as a chip or an IC of combination-type. For example, the semiconductor die 130 may be a WiFi chip simultaneously including both of a RF chip and a digital chip. The disclosure is not limited thereto.

In alternative embodiments, the semiconductor die 130 may further include additional semiconductor die(s) of the same type or different types. For example, the additional semiconductor die(s) may include digital chips, analog chips or mixed signal chips, such as ASIC chips, sensor chips, wireless and RF chips, memory chips, logic chips or voltage regulator chips. The disclosure is not limited thereto.

As shown in FIG. 2, for example, positioning locations of the conductive pillars 120 are located aside of a positioning location of the semiconductor die 130 on the X-Y plane. In some embodiments, along the direction Z, a height of the conductive pillars 120 is greater than a height of the semi-

conductor die 130; however, the disclosure is not limited thereto. In an alternative embodiment, the height of the conductive pillars 120 may be less than or substantially equal to the height of the semiconductor die 130. In one embodiment, the conductive pillars 120 may be formed prior to the formation of the semiconductor die 130; however, the disclosure is not limited thereto. In an alternative embodiment, the conductive pillars 120 may be formed after the formation of the semiconductor die 130.

Referring to FIG. 3, in some embodiments, an insulating encapsulation 140a is formed over the carrier C (e.g., on the solder resist layer 110a) to encapsulate the conductive pillars 120 and the semiconductor die 130. In other words, the insulating encapsulation 140a is formed on the solder resist layer 110a, the conductive pillars 120 and the semiconductor die 130, where the conductive pillars 120 and the semiconductor die 130 (disposed with the connecting film DA) are covered by and embedded in the insulating encapsulation 140a. As shown in FIG. 3, for example, the insulating encapsulation 140a at least fills up the gaps between the conductive pillars 120 and the gaps between the conductive pillars 120, the semiconductor die 130 and the connecting films DA. In some embodiments, sidewalls 120s of the conductive pillars 120 and sidewalls 130s of the semiconductor die 130 are covered by the insulating encapsulation 140a. In some embodiments, the surface S1 of the solder resist layer 110a exposed by the conductive pillars 120 and the semiconductor die 130 are covered by the insulating encapsulation 140a. For example, as shown in FIG. 3, the solder resist layer 110a, the conductive pillars 120, the semiconductor die 130 and the connecting film DA are not accessibly revealed by the insulating encapsulation 140a.

In some embodiments, the insulating encapsulation 140a is a molding compound formed by a molding process. In some embodiments, the insulating encapsulation 140a, for example, may include polymers (such as epoxy resins, phenolic resins, silicon-containing resins, or other suitable resins), dielectric materials having low permittivity and low loss tangent properties, or other suitable materials. The disclosure is not limited thereto. In an alternative embodiment, the insulating encapsulation 140a may include an acceptable insulating encapsulation material. In some embodiments, the insulating encapsulation 140a may further include inorganic filler or inorganic compound (e.g. silica, clay, and so on) which can be added therein to optimize the CTE of the insulating encapsulation 140a. In the disclosure, the material of the insulating encapsulation 140a is different from the material of the solder resist layer 110a, where the CTE of the insulating encapsulation 140a is less than the CTE of the solder resist layer 110a.

Referring to FIG. 4, in some embodiments, the insulating encapsulation 140a is planarized to form an insulating encapsulation 140 exposing the conductive pillars 120 and the semiconductor die 130. In certain embodiments, as shown in FIG. 4, after the planarization, top surfaces 120t of the conductive pillars 120 and a top (or front) surface 130t of the semiconductor die 130 (e.g. top surfaces (not labelled) of the connecting vias 135 and the protection layer 136 of the semiconductor die 130) are exposed by a top surface 140t of the insulating encapsulation 140. That is, for example, the top surface 130t of the semiconductor die 130 and the top surfaces 120t of the conductive pillars 120 become substantially leveled with the top surface 140t of the insulating encapsulation 140. In other words, the top surface 130t of the semiconductor die 130, the top surfaces 120t of the conductive pillars 120, and the top surface 140t of the insulating encapsulation 140 are substantially coplanar to

each other. In some embodiments, the conductive pillars 120 each penetrate through the insulating encapsulation 140 and have the top surfaces 120t exposed therefrom, while the semiconductor die 130 are embedded inside the insulating encapsulation 140 and has the top surface 130t exposed therefrom. For example, as shown in FIG. 4, the conductive pillars 120 and the semiconductor die 130 are accessibly revealed by the insulating encapsulation 140.

The insulating encapsulation 140a may be planarized by mechanical grinding or chemical mechanical polishing (CMP), for example. After the planarizing step, a cleaning step may be optionally performed, for example to clean and remove the residue generated from the planarizing step. However, the disclosure is not limited thereto, and the planarizing step may be performed through any other suitable method.

In some embodiments, during planarizing the insulating encapsulation 140a, the connecting vias 135 and the protection layer 136 of the semiconductor die 130 and the conductive pillars 120 may also be planarized. In certain embodiments, the planarizing step may, for example, be performed on the over-molded insulating encapsulation 140a to level the top surface 140t of the insulating encapsulation 140, the top surfaces 120t of the conductive pillars 120 and the top surface 130t of the semiconductor die 130.

Referring to FIG. 5, in some embodiments, a redistribution circuit structure 150 is formed on the conductive pillars 120, the semiconductor die 130, and the insulating encapsulation 140. As shown in FIG. 5, the redistribution circuit structure 150 is directly formed on the top surfaces 120t of the conductive pillars 120, the top surface 130t of the semiconductor die 130, and the top surface 140t of the insulating encapsulation 140, for example. In some embodiments, the redistribution circuit structure 150 is electrically connected to the conductive pillars 120, and is electrically connected to the semiconductor die 130 through the connecting vias 135. In some embodiments, through the redistribution circuit structure 150, the semiconductor die 130 is electrically connected to the conductive pillars 120. In alternative embodiments of which more than one semiconductor dies 130 are included, the semiconductor dies 130 are electrically communicated through the redistribution circuit structure 150. As shown in FIG. 5, for example, the redistribution circuit structure 150 is referred to as a front-side redistribution layer of the semiconductor die 130.

For example, as shown in FIG. 5, along the stacking direction (e.g. the direction Z), the semiconductor die 130 is located between the redistribution circuit structure 150 and the connecting film DA. In addition, a portion of each of the conductive pillars 120 is located between the redistribution circuit structure 150 and the solder resist layer 110a, and other portion of each of the conductive pillars 120 is located between the redistribution circuit structure 150 and the debond layer DB. The insulating encapsulation 140 is located between the redistribution circuit structure 150 and the solder resist layer 110a, for example.

In some embodiments, the formation of the redistribution circuit structure 150 includes sequentially forming one or more dielectric layers 152 and one or more metallization layers 154 in alternation. For example, as shown in FIG. 5, the redistribution circuit structure 150 includes dielectric layers 152a, 152b, 152c, 152d and the metallization layers 154a, 154b, 154c. In some embodiments, the metallization layer 154a is sandwiched between the dielectric layers 152a and 152b, the metallization layer 154b is sandwiched between the dielectric layers 152b and 152c, the metallization layer 154c is sandwiched between the dielectric layers

152c and **152d**. The disclosure is not limited thereto. It should be noted that the redistribution circuit structure **150** is not limited to include four dielectric layers and three metallization layers. For example, the number of the metallization layers and the numbers of the dielectric layers may be one or more than one.

In some embodiments, the material of the dielectric layers **152** may be PI, PBO, BCB, a nitride such as silicon nitride, an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), a combination thereof or the like, which may be patterned using a photolithography and/or etching process. In some embodiments, the material of the dielectric layers **152** formed by suitable fabrication techniques such as spin-on coating, chemical vapor deposition (CVD), plasma-enhanced chemical vapor deposition (PECVD) or the like. The disclosure is not limited thereto. In some embodiments, the material of the metallization layers **154** may be made of conductive materials formed by electroplating or deposition, such as aluminum, titanium, copper, nickel, tungsten, and/or alloys thereof, which may be patterned using a photolithography and etching process. In some embodiments, the metallization layers **154** may be patterned copper layers or other suitable patterned metal layers.

In some embodiments, a seed layer (not shown) may be formed between one metallization layer **154** and a respective one dielectric layer **152** underlying thereto. In some embodiments, the seed layer may be referred to as a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, a material of the seed layer may include titanium, copper, molybdenum, tungsten, titanium nitride, titanium tungsten, combinations thereof, or the like. For example, the seed layer may include a titanium layer and a copper layer over the titanium layer. In some embodiments, the seed layer may be formed using, for example, sputtering, physical vapor deposition (PVD), or the like.

In some embodiments, portions of a top surface of a topmost layer (e.g. the metallization layer **154c**) of the metallization layers **154** are exposed by a topmost layer (e.g. the dielectric layer **152d**) of the dielectric layers **152** to electrically connect overlying conductive features (e.g. later-formed under bump metallurgy (UBM) patterns **162** and/or contact pads **164**). For example, as shown in FIG. 5, the portions of the top surface of the metallization layer **154c** are exposed by openings OP2 formed in the dielectric layer **152d**. In some embodiments, portions of a bottom surface of a lowest layer (e.g. the metallization layer **154a**) of the metallization layers **154** are exposed by a lowest layer (e.g. the dielectric layer **152a**) of the dielectric layers **152** to electrically connect underlying conductive features (e.g. the conductive pillars **120** and the connecting vias **135** of the semiconductor die **130**). As shown in FIG. 5, in some embodiments, the conductive pillars **120**, and the redistribution circuit structure **150** provide a routing function for the semiconductor die **130**.

Referring to FIG. 6, in some embodiments, a plurality of UBM patterns **162** are formed to be disposed on the exposed top surfaces of the topmost layer (e.g. the metallization layer **154c**) of the metallization layers **154** for electrically connecting with conductive elements (e.g. conductive balls or conductive bumps). In some embodiments, prior to, during, or after the formation of the UBM patterns **162**, a plurality of contact pads **164** are optionally formed to be disposed on some of the exposed top surfaces of the topmost layer (e.g. the metallization layer **154c**) of the metallization layers **154** for electrically connecting with semiconductor elements

(e.g. semiconductor active or passive devices). The number of the UBM patterns **162** and the number of the contact pads **164** are not limited as depicted in the disclosure, and may be selected and designated based on the demand and design layout, the disclosure is not limited thereto.

For example, as shown in FIG. 6, the UBM patterns **162** and the contact pads **164** are formed on and electrically connected to the redistribution circuit structure **150**. For example, the UBM patterns **162** and the contact pads **164** are disposed on the dielectric layer **152d** and further in contact with the portions of the metallization layer **154c** exposed by the openings OP2 formed in the dielectric layer **152d**. In some embodiments, the materials of the UBM patterns **162** and the contact pads **164** may include copper, nickel, titanium, tungsten, or alloys thereof or the like, and may be formed by an electroplating process, for example. In one embodiment, the material of the UBM patterns **162** may be the same as that of the contact pads **164**. In an alternative embodiment, the material of the UBM patterns **162** may be different from that of the contact pads **164**. In one embodiment, there may be only the UBM patterns **162**; however, the disclosure is not limited thereto. In one embodiment, the UBM patterns **162** and the contact pads **164** may be formed in the same processing step. In an alternative embodiment, the UBM patterns **162** and the contact pads **164** may be formed in different processing steps.

Continued on FIG. 6, in some embodiments, a plurality of conductive elements **172** are formed on the redistribution circuit structure **150**. For example, the conductive elements **172** are disposed on the UBM patterns **162** located on the redistribution circuit structure **150**. In some embodiments, the conductive elements **172** may be disposed on the UBM patterns **162** by ball placement process or reflow process. In some embodiments, the conductive elements **172** are, for example, controlled collapse chip connection (C4) bumps, ball grid array (BGA) balls, solder balls/bumps or other connectors. When solder is used, the solder may include either eutectic solder or non-eutectic solder. The solder may include lead or be lead-free, and may include Sn—Ag, Sn—Cu, Sn—Ag—Cu, or the like. In the disclosure, for one embodiment, the conductive elements **172** may be referred to as conductive connectors for connecting with another package; or for another embodiment, the conductive elements **172** may be referred to as conductive terminals for inputting/outputting electric and/or power signals. In some embodiments, the conductive elements **172** are electrically connected (e.g. electrically coupled) to the redistribution circuit structure **150** through the UBM patterns **162**. As shown in the FIG. 6, some of the conductive elements **172** are electrically connected to the semiconductor die **130** through the UBM patterns **162** and the redistribution circuit structure **150**, and some of the conductive elements **172** are electrically connected to the conductive pillars **120** through the UBM patterns **162** and the redistribution circuit structure **150**, for example. The number of the conductive elements **172** is not limited to the disclosure, and may be designated and selected based on the number of the UBM patterns **162**.

In some embodiments, one or more semiconductor devices **174** are provided and disposed on the redistribution circuit structure **150**. For example, the semiconductor devices **174** are disposed on the contact pads **164**, and are electrically connected to the redistribution circuit structure **150** through the contact pads **164**. In some embodiments, some of the semiconductor devices **174** are electrically connected to the semiconductor die **130** through the contact pads **164** and the redistribution circuit structure **150**. In some embodiments, some of the semiconductor devices **174** are

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electrically connected to the conductive pillars **120** through the contact pads **164** and the redistribution circuit structure **150**. In some embodiments, some of the semiconductor devices **174** are electrically connected to at least one of the conductive elements **172** through the contact pads **164**, the redistribution circuit structure **150** and the UBM patterns **162**. In some embodiments, the semiconductor devices **174** may be disposed on the contact pads **164** through reflow process or flip chip bonding. In some embodiments, the semiconductor devices **174** include surface mount devices (e.g. passive devices, such as, capacitors, resistors, inductors, combinations thereof, or the like). The number of the semiconductor devices **174** can be selected based on the number of the contact pads **164**. In an alternative embodiment, the semiconductor devices **174** may include surface mount devices of the same type or different types, the disclosure is not limited thereto. In alternative embodiments, the semiconductor devices **174** are optional, and may be omitted.

In some embodiments, along the direction Z, the conductive elements **172** and the semiconductor devices **174** are formed on a side of the redistribution circuit structure **150**, and the insulating encapsulation **140** is formed on other side of the redistribution circuit structure **150**. That is, the redistribution circuit structure **150** is located between the insulating encapsulation **140** and the conductive elements **172** and between the insulating encapsulation **140** and the semiconductor devices **174**. In some embodiments, the semiconductor devices **174** may be formed prior to the formation of the conductive elements **172**. In an alternative embodiment, the conductive elements **172** may be formed prior to the formation of the semiconductor devices **174**. The disclosure is not limited to thereto.

Referring to FIG. 7, in some embodiments, the whole structure depicted in FIG. 6 along with the carrier C is flipped (turned upside down), where the conductive elements **172** are placed to a holding device HD, and the carrier C is then debonded from the solder resist layer **110a**. In some embodiments, the holding device HD may be an adhesive tape, a carrier film or a suction pad. The solder resist layer **110a** is easily separated from the carrier C due to the debond layer DB. In some embodiments, the carrier C is detached from the solder resist layer **110a** through a debonding process, and the carrier C and the debond layer DB are removed. For example, the surface S2 of the solder resist layer **110a** and bottom surface **120b** of the conductive pillars **120** are exposed. In one embodiment, the debonding process is a laser debonding process. During the debonding step, the holding device HD is used to secure the package depicted in FIG. 6 before debonding the carrier C and the debond layer DB. As shown in FIG. 7, for example, the openings OP1 are filled up with the conductive pillars **120**, where the surfaces **120b** of the conductive pillars **120** are substantially coplanar with the surface S2 of the solder resist layer **110a**.

Referring to FIG. 8, in some embodiments, a plurality of conductive elements **180a** are formed on the bottom surfaces **120b** of the conductive pillars **120**. For example, the bottom surfaces **120b** of the conductive pillars **120** exposed by the surface S2 of the solder resist layer **110a** are covered by the conductive elements **180a**. For example, the conductive elements **180a** include conductive bumps or conductive balls. The conductive elements **180a** may be pre-solder pastes, for example. In an alternative embodiment, the conductive elements **180a** may be pre-solder blocks. In some embodiments, the material of the conductive elements **180a** may include a lead-free solder material (such as Sn—Ag base or Sn—Ag—Cu base materials) with or with-

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out additional impurity (such as Ni, Bi, Sb, Au, or the like). The disclosure is not limited thereto. In the disclosure, the conductive elements **180a** may also be referred to as conductive terminals for electrical connection to external elements (e.g. an additional semiconductor package/device, a circuit substrate, etc.). As shown in FIG. 8, the conductive elements **180a** are formed outside of the openings OP1 and covered the surfaces **120b** of the conductive pillars **120**. That is, the conductive elements **180a** are rest at the surface S2 of the solder resist layer **110a** and are protruding outwards from the surface S2.

In some embodiments, the conductive elements **172** are released from the holding device HD to form the package structure **10a**. In some embodiments, a dicing (singulating) process is performed to cut a plurality of the package structures **10a** interconnected therebetween into individual and separated package structures **10a** before releasing the conductive elements **172** from the holding device HD. In one embodiment, the dicing process is a wafer dicing process including mechanical blade sawing or laser cutting. Up to here, the manufacture of the package structure **10a** is completed. The package structure **10a** depicted in FIG. 8 may be referred to as an integrated fan-out (semiconductor) package structure having dual-side terminals.

In some alternative embodiments, the package structure **10a** may be further mounted with a circuit substrate, an interposer, an additional package, chips/dies or other electronic devices to form a stacked package structure or a package on package (PoP) structure through the conductive elements **172** and/or other the conductive elements **180a** based on the design layout and the demand.

FIG. 9 is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure. The elements similar to or substantially the same as the elements described previously will use the same reference numbers, and certain details or descriptions (e.g. the materials, formation processes, positioning configurations, etc.) of the same elements would not be repeated herein. Referring to FIG. 9, for example, a semiconductor package **800** is provided and then bonded to the package structure **10a**, thereby forming a package structure SP1 of a stacked structure. The detail of the package structure **10a** is described in FIG. 8, and thus is not repeated herein for simplicity.

In some embodiments, the semiconductor package **800** has a substrate **810**, semiconductor dies **820a** and **820b**, bonding wires **830a** and **830b**, conductive pads **840**, conductive pads **850**, an insulating encapsulation **860**, and conductive elements **870**. For example, the semiconductor die **820a** and the semiconductor die **820b** are provided and disposed on the substrate **810**. In some embodiments, the connecting film DA2 is located between the semiconductor die **820a** and the substrate **810**, and the connecting film DA3 is located between the semiconductor die **820a** and the semiconductor die **820b**. In some embodiments, due to the connecting films DA2 and DA3 respectively provided between the semiconductor die **820a** and the substrate **810** and between the semiconductor dies **820a** and **820b**, the semiconductor dies **820a**, **820b** are stably adhered to the substrate **810**. In some embodiments, the connecting films DA2, DA3 may be, for example, a die attach film, a layer made of adhesives or epoxy resin, or the like.

For example, the semiconductor dies **820a** and **820b** are mounted on one surface (e.g. a surface **810a**) of the substrate **810**. In some embodiments, the semiconductor dies **820a** and **820b** may be logic chips (e.g., central processing units, microcontrollers, etc.), memory chips (e.g., dynamic ran-

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dom access memory (DRAM) chips, static random access memory (SRAM) chips, etc.), power management chips (e.g., power management integrated circuit (PMIC) chips), radio frequency (RF) chips, sensor chips, signal processing chips (e.g., digital signal processing (DSP) chips), and/or front-end chips (e.g., analog front-end (AFE) chips, the like, or a combination thereof). The semiconductor dies **820a** and **820b** are DRAM chips, as shown in FIG. 9, for example. In one embodiment, the semiconductor dies **820a** and **820b** may be the same. However, the disclosure is not limited thereto; in an alternative embodiment, the semiconductor dies **820a** and **820b** may be different from each other.

In some embodiments, the bonding wires **830a** and **830b** are respectively used to provide electrical connections between the semiconductor dies **820a**, **820b** and some of the conductive pads **840** (such as bonding pads) located on the surface **810a** of the substrate **810**. Owing to the bonding wires **830a** and **830b**, the semiconductor dies **820a** and **820b** are electrically connected to the substrate **810**.

In some embodiments, the insulating encapsulation **860** is formed on the surface **810a** of the substrate **810** to encapsulate the semiconductor dies **820a**, **820b**, the bonding wires **830a**, **830b**, and the conductive pads **840** to protect these components. In some embodiments, the material of the insulating encapsulation **860** is the same as the materials of the insulating encapsulation **140/140a**, and thus is not repeated herein for simplicity. In one embodiment, the material of the insulating encapsulation **860** is different from the materials of the insulating encapsulation **140/140a**, the disclosure is not limited thereto.

In some embodiments, interconnects (not shown) or through insulator vias (not shown) embedded in the substrate **810** may be used to provide electrical connection between the conductive pads **840** and the conductive pads **850** (such as bonding pads) that are located on another surface (e.g. a surface **810b** opposite to the surface **810a** along the direction Z) of the substrate **810**. In certain embodiments, some of the conductive pads **850** are electrically connected to the semiconductor dies **820a** and **820b** through these insulator vias or interconnects (not shown) in addition to some of the conductive pads **840** and the bonding wires **830a**, **830b**.

In some embodiments, conductive elements **870** are disposed on the conductive pads **850** and over the surface **810b** of the substrate **810**. The formation and material of the conductive elements **870** may be the same or similar to the formation and material of the conductive elements **170** or the formation and material of the conductive elements **180a**, and thus are not repeated herein for simplicity. As shown in FIG. 9, for example, the conductive elements **180a** of the package structure **10a** is bonded to the conductive elements **870** of the semiconductor package **800**, and thus joints **300a** are formed, in the package structure SP1, to electrically connect the semiconductor die **130** to the semiconductor dies **820a**, **820b**. In some embodiments, the joints **300a** are located between the solder resist layer **110a** and the substrate **810**. In other words, the semiconductor dies **820a**, **820b** of the semiconductor package **800** are electrically connected to the semiconductor die **130** of the package structure **10a** through the bonding wires **830a** and **830b**, the conductive pads **840**, **850** disposed on the substrate **810**, the joints **300a** (including the conductive elements **870** and **180a**), the conductive pillars **120**, and the redistribution circuit structure **150**.

In addition, an underfill (not shown) may be optimally formed to wrap around sidewalls of the joints **300a**. The underfill may further fill up the gaps between the solder

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resist layer **110a** of the package structure **10a** and the substrate **810** of the semiconductor package **800**, for example. The underfill may be any acceptable material, such as a polymer, epoxy, molding underfill, or the like, for example. In one embodiment, the underfill may be formed by underfill dispensing or any other suitable method. Owing to the underfill, the bonding strength between the package structure **10a** and the semiconductor package **800** are enhanced, thereby improving the reliability of the package structure SP1 depicted FIG. 9.

FIG. 10 is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure. FIG. 11 is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure. Referring to FIG. 8 and FIG. 10 together, the semiconductor package structure **10a** depicted in FIG. 8 and a semiconductor package structure **10b** depicted in FIG. 10 are similar, the difference is that an additional element, e.g. a redistribution circuit structures **190**, is further included in the semiconductor package structure **10b**. The elements similar to or substantially the same as the elements described previously will use the same reference numbers, and certain details or descriptions (e.g. the materials, formation processes, positioning configurations, etc.) of the same elements would not be repeated herein. With the embodiments of which the redistribution circuit structure **190** is included, the redistribution circuit structure **190** is formed prior to the formation of the conductive pillars **120** and after the formation of the solder resist layer **110a**.

In some embodiments, the redistribution circuit structure **190** is located on the surface S1 of the solder resist layer **110a** and a bottom surface **140b** of the insulating encapsulation **140**. Along the direction Z, the bottom surface **140b** of the insulating encapsulation **140** is opposite to the top surface **140t** of the insulating encapsulation **140**. In some embodiments, the redistribution circuit structure **190** is electrically connected to the conductive pillars **120**, is electrically connected to the redistribution circuit structure **150** through the conductive pillars **120**, and is electrically connected to the semiconductor die **130** through the conductive pillars **120**, the redistribution circuit structure **150**, and the connecting vias **135**. In some embodiments, through the conductive pillars **120**, the redistribution circuit structure **150** and the UBM patterns **162**, the redistribution circuit structure **190** is further electrically connected to at least one of the conductive elements **172**. In some embodiments, through the conductive pillars **120**, the redistribution circuit structure **150** and the contact pads **164**, the redistribution circuit structure **190** is further electrically connected to at least one of the semiconductor devices **174**. In some embodiments, the redistribution circuit structure **190** is electrically connected to the conductive elements **180a**. In such embodiments, the semiconductor die **130** is electrically connected to at least some of the conductive elements **180a** through the redistribution circuit structure **150**, the conductive pillars **120** and the redistribution circuit structure **190**. As shown in FIG. 10, for example, the redistribution circuit structure **190** is referred to as a back-side redistribution layer of the semiconductor die **130**.

In some embodiments, the formation of the redistribution circuit structure **190** includes sequentially forming one or more dielectric layers **192** and one or more metallization layers **194** in alternation. For illustrative purposes, as shown in FIG. 10, the redistribution circuit structure **190** includes one dielectric layer **192** and one metallization layer **194**. It is appreciated that the redistribution circuit structure **190** is not limited to include one dielectric layer **192** and one

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metallization layer 194. The number of the dielectric layer 192 and the number of the metallization layer 194 may be more than one based on the demand and the design layout. In some embodiments, the metallization layer 194 is located on the solder resist layer 110a; the dielectric layer 192 is located on the metallization layer 194; and the conductive pillars 120, the semiconductor die 130 and the insulating encapsulation 140 are located on the dielectric layer 192. For example, as shown in FIG. 10, the conductive pillars 120 individually penetrate the dielectric layer 192 to electrically connect to the metallization layer 194, and the metallization layer 194 penetrates the solder resist layer 110a to electrically connect to the conductive elements 180a. That is, the conductive elements 180a are, via the redistribution circuit structure 190, electrically connected to the conductive pillars 120.

In some embodiments, the material and formation of the dielectric layer 192 may be the same as the material and formation of the dielectric layers 152, and the material and formation of the metallization layer 194 may be the same as the material and formation of the metallization layers 154, thus is not repeated herein. In an alternative embodiment, the material of the dielectric layer 192 may be the same as or different from the material of the dielectric layers 152. In an alternative embodiment, the material of the metallization layer 194 may be the same as or different from the material of the metallization layers 154. The disclosure is not limited thereto. In some alternative embodiments, a seed layer (not shown) may be formed between the metallization layer 194 and the solder resist layer 110a underlying thereto. In some embodiments, the seed layer may be referred to as a metal layer, which may be a single layer or a composite layer comprising a plurality of sub-layers formed of different materials. In some embodiments, a material of the seed layer may include titanium, copper, molybdenum, tungsten, titanium nitride, titanium tungsten, combinations thereof, or the like. For example, the seed layer may include a titanium layer and a copper layer over the titanium layer. In some embodiments, the seed layer may be formed using, for example, sputtering, physical vapor deposition (PVD), or the like.

In alternative embodiments, the package structure 10a in the package structure SP1 may be replaced with the package structure 10b of FIG. 10, see a package structure SP2 depicted in FIG. 11. In some embodiments, as shown in FIG. 11, for the package structure SP2, the semiconductor package 800 is bonded to the semiconductor package structure 10b depicted in FIG. 10 by connecting the conductive elements 870 and the conductive elements 180a (e.g. forming the joints 300a). In such embodiments, the semiconductor dies 820a, 820b of the semiconductor package 800 are electrically connected to the semiconductor die 130 of the package structure 10b through the bonding wires 830a and 830b, the conductive pads 840 and 850 disposed on the substrate 810, the joints 300a (including the conductive elements 870 and 180a), the redistribution circuit structure 190, the conductive pillars 120, and the redistribution circuit structure 150.

FIG. 12 to FIG. 14 are schematic cross sectional views of various stages in a manufacturing method of a package structure in accordance with some embodiments of the disclosure. FIG. 15 is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure. The elements similar to or substantially the same as the elements described previously will use the same reference numbers, and certain details or descriptions (e.g.

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the materials, formation processes, positioning configurations, etc.) of the same elements would not be repeated herein.

Referring in FIG. 12, in some embodiments, a carrier C with a debond layer DB and a solder resist layer 110b coated thereon is provided. For example, the debond layer DB is located between the carrier C and the solder resist layer 110b. The material of the carrier C and the formation and material of the debond layer DB have been described in FIG. 1, and thus are not repeating herein for simplicity. In the disclosure, the formation and material of the solder resist layer 110b is similar to the formation and the material of the solder resist layer 110a described in FIG. 1, however, the solder resist layer 110b is non-photosensitive. In some embodiments, a thickness T110b of the solder resist layer 110b is approximately from 10 μm to 30 μm . Due to the solder resist layer 110b (e.g. the low CTE value) in addition to its specific thickness range, better warpage control (e.g., warpage being less than or substantially equal to 80 μm at room temperature and being greater than or substantially equal to -80 μm) to the package structure 10c is achieved.

Thereafter, in some embodiments, at least one conductive pillar 120 and at least one semiconductor die 130 are formed on the solder resist layer 110b (e.g. on a surface S3 of the solder resist layer 110b). For illustrative purposes, the at least one conductive pillar 120 include a plurality of conductive pillars 120 (e.g. two conductive pillars 120), and at least one semiconductor die 130 include one semiconductor die 130; however, the disclosure is not limited thereto. The formation and material of the conductive pillars 120 and the formation and material of the semiconductor die 130 have been described in FIG. 1, and thus are not repeating herein for simplicity. In some embodiments, the conductive pillars 120 and the semiconductor die 130 are arranged side-by-side on the solder resist layer 110b along the X-Y plane. Due to the solder resist layer 110b is non-photosensitive, no opening OP1 is formed in the solder resist layer 110b prior to the formations of conductive pillars 120 and the semiconductor die 130.

Referring to FIG. 13, in some embodiments, the previously described manufacturing process as described in FIG. 3 to FIG. 7 and a subsequent patterning process are performed on the structure depicted in FIG. 12. For example, by performing the previously described manufacturing process as described in FIG. 3 to FIG. 7 on the structure depicted in FIG. 12, the carrier C and the debond layer DB are removed from the solder resist layer 110b, where a surface S4 of the solder resist layer 110b is exposed. The surface S4 is opposite to the surface S3 along the direction Z.

Thereafter, in some embodiments, the exposed solder resist layer 110b is then patterned to form a plurality of openings OP3 therein. The patterning process, for example, includes a laser drill process. For example, only two openings OP3 are shown in FIG. 13 for illustrative purposes; however, the number of the openings OP3 may be more than two based on the demand and the design layout. Additionally, on the X-Y plane, dimensions (e.g. maximum widths) of the openings OP3 may be the same, however the disclosure is not limited thereto. In an alternative embodiment, according to the design layout and/or demand, the dimensions of the openings OP3 may be the different from each other or may be different in a manner of different groups. In one embodiment, on the X-Y plane, a cross-sectional shape of the openings OP3 individually may be round, elliptical, oval, tetragonal, octagonal or any suitable polygonal shape; the disclosure is not limited thereto. As illustrated in FIG. 13, after patterning the solder resist

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layer **110b**, the bottom surfaces **120b** of the conductive pillars **120** are exposed by the openings **OP3**. As shown in FIG. **13**, for example, the openings **OP3** are not filled with the conductive pillars **120**, where the surfaces **120b** of the conductive pillars **120** are substantially coplanar with the surface **S3** of the solder resist layer **110b**. That is, the conductive pillars **120** are not extended into the openings **OP3**. In the solder resist layer **110b** shown in FIG. **13**, an angle $\theta 3$ between the surface **S3** of the solder resist layer **110b** and a sidewall **SW2** of each opening **OP3** is approximately 40 degrees to 60 degrees, and an angle $\theta 4$ (i.e. $\theta 4 = 180 \text{ degrees} - \theta 3$) between the surface **S4** of the solder resist layer **110b** and the sidewall **SW2** of each opening **OP3** is 120 degrees to 140 degrees. With the formation of the solder resist layer **110b** having the openings **OP3**, the manufacturing cost is further reduced.

Referring to FIG. **14**, in some embodiments, a plurality of conductive elements **180b** are formed on the bottom surfaces **120b** of the conductive pillars **120** and in the openings **OP3**. For example, the bottom surfaces **120b** of the conductive pillars **120** exposed by the solder resist layer **110a** are covered by the conductive elements **180b**. The formation and material of the conductive elements **180b** is similar to the formation and material of the conductive elements **180a** described in FIG. **8**, and thus are not repeated herein. In the disclosure, the conductive elements **180b** may also be referred to as conductive terminals for electrical connection to external elements (e.g. an additional semiconductor package/device, a circuit substrate, etc.). As shown in FIG. **14**, the conductive elements **180b** are formed inside of the openings **OP3** and covered the surfaces **120b** of the conductive pillars **120**. That is, the conductive elements **180b** are rest at a first plane where the surface **S3** of the solder resist layer **110a** located at and are protruding from the first plane towards a second plane where the surface **S4** located at.

In some embodiments, after forming the conductive elements **180b**, the conductive elements **172** are released from the holding device **HD** to form the package structure **10c**. In some embodiments, a dicing (singulating) process is performed to cut a plurality of the package structures **10c** interconnected therebetween into individual and separated package structures **10c** before releasing the conductive elements **172** from the holding device **HD**. In one embodiment, the dicing process is a wafer dicing process including mechanical blade sawing or laser cutting. Up to here, the manufacture of the package structure **10c** is completed. The package structure **10c** depicted in FIG. **14** may be referred to as an integrated fan-out (semiconductor) package structure having dual-side terminals.

In some alternative embodiments, the package structure **10c** may be further mounted with a circuit substrate, an interposer, an additional package, chips/dies or other electronic devices to form a stacked package structure or a package on package (PoP) structure through the conductive elements **172** and/or other the conductive elements **180b** based on the design layout and the demand. For example, the package structure **10c** of FIG. **14** is bonded to a semiconductor package **800** to form a package structure **SP3**, as shown in FIG. **15**.

Referring to FIG. **15**, for the package structure **SP3**, the semiconductor package **800** is bonded to the semiconductor package structure **10c** depicted in FIG. **14** by connecting the conductive elements **870** and the conductive elements **180b** (e.g. forming the joints **300b**), for example. In some embodiments, a portion of each of the joints **300b** is located in the openings **OP3**. In such embodiments, the semiconductor

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dies **820a**, **820b** of the semiconductor package **800** are electrically connected to the semiconductor die **130** of the package structure **10c** through the bonding wires **830a** and **830b**, the conductive pads **840** and **850** disposed on the substrate **810**, the joints **300b** (including the conductive elements **870** and **180b**), the conductive pillars **120**, and the redistribution circuit structure **150**.

FIG. **16** is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure. FIG. **17** is a schematic cross sectional view of a package structure in accordance with some embodiments of the disclosure. Referring to FIG. **14** and FIG. **16** together, the semiconductor package structure **10c** depicted in FIG. **14** and a semiconductor package structure **10d** depicted in FIG. **16** are similar, the difference is that an additional element, e.g. a redistribution circuit structures **190**, is further included in the semiconductor package structure **10d**. The elements similar to or substantially the same as the elements described previously will use the same reference numbers, and certain details or descriptions (e.g. the materials, formation processes, positioning configurations, etc.) of the same elements would not be repeated herein.

With the embodiments of which the redistribution circuit structure **190** is included, the redistribution circuit structure **190** is formed prior to the formation of the conductive pillars **120** and after the formation of the solder resist layer **110b**. The formation and material of the redistribution circuit structure **190** has been described in FIG. **10**, and thus are not repeating herein for simplicity. In some embodiments, the redistribution circuit structure **190** is located on the surface **S3** of the solder resist layer **110b** and a bottom surface **140b** of the insulating encapsulation **140**. Similar to the redistribution circuit structure **190** of the package structure **10b** described in FIG. **10** that providing routing function, for example, the redistribution circuit structure **190** is also referred to as a back-side redistribution layer of the semiconductor die **130** of the package structure **10d** depicted in FIG. **16**. That is, the redistribution circuit structure **190** also provided routing function to the semiconductor die **130** with other conductive components inside the package structure **10d** similar to the redistribution circuit structure **190** of the package structure **10b**.

In some alternative embodiments, the package structure **10d** may be further mounted with a circuit substrate, an interposer, an additional package, chips/dies or other electronic devices to form a stacked package structure or a package on package (PoP) structure through the conductive elements **172** and/or other the conductive elements **180b** based on the design layout and the demand. For example, the package structure **10d** of FIG. **16** is bonded to a semiconductor package **800** to form a package structure **SP4**, as shown in FIG. **17**. The detail of the semiconductor package **800** is described in FIG. **9**, and thus is not repeated herein. In some embodiments, as shown in FIG. **17**, for the package structure **SP4**, the semiconductor package **800** is bonded to the semiconductor package structure **10d** depicted in FIG. **16** by connecting the conductive elements **870** and the conductive elements **180b** (e.g. forming the joints **300b**). In such embodiments, the semiconductor dies **820a**, **820b** of the semiconductor package **800** are electrically connected to the semiconductor die **130** of the package structure **10d** through the bonding wires **830a** and **830b**, the conductive pads **840** and **850** disposed on the substrate **810**, the joints **300b** (including the conductive elements **870** and **180b**), the redistribution circuit structure **190**, the conductive pillars **120**, and the redistribution circuit structure **150**.

In accordance with some embodiments, a package structure includes a semiconductor die, conductive pillars, an insulating encapsulation, a redistribution circuit structure, and a solder resist layer. The conductive pillars are arranged aside of the semiconductor die. The insulating encapsulation encapsulates the semiconductor die and the conductive pillars, and the insulating encapsulation has a first surface and a second surface opposite to the first surface. The redistribution circuit structure is located on the first surface of the insulating encapsulation. The solder resist layer is located on the second surface of the insulating encapsulation, wherein a material of the solder resist layer includes a filler.

In accordance with some embodiments, a package structure includes a semiconductor die, an insulating encapsulation, a first redistribution circuit structure, and a solder resist layer. The semiconductor die has a front surface and a rear surface opposite to the front surface. The insulating encapsulation laterally encapsulates the semiconductor die. The first redistribution circuit structure is located on the front surface of the semiconductor die. The solder resist layer is located over the rear surface of the semiconductor die, wherein a material of the solder resist layer comprises a filler.

In accordance with some embodiments, a method of manufacturing package structure includes the following steps, providing a carrier; forming a solder resist layer with a material comprising a filler on the carrier; disposing a semiconductor die and conductive pillars on the solder resist layer; encapsulating the semiconductor die and the conductive pillars in the insulating encapsulation; forming a first redistribution circuit structure on the insulating encapsulation; disposing first conductive element over the conductive pillars, the conductive pillars being between the first redistribution circuit structure and the first conductive elements; and disposing second conductive elements on the first redistribution circuit structure, the first redistribution circuit structure being between the insulating encapsulation and the second conductive elements.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the disclosure. Those skilled in the art should appreciate that they may readily use the disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the disclosure.

What is claimed is:

1. A package structure, comprising:

a semiconductor die;

conductive pillars, next to and electrically coupled to the semiconductor die;

an insulating encapsulation, laterally encapsulating the semiconductor die and the conductive pillars, the insulating encapsulation having a first surface and a second surface opposite to the first surface;

a redistribution circuit structure, disposed on the first surface of the insulating encapsulation and electrically coupled to the semiconductor die;

a solder resist layer, disposed on the second surface of the insulating encapsulation, wherein the solder resist layer is a single-layer structure comprising a plurality of openings exposing surfaces of the conductive pillars;

first terminals, disposed on and electrically coupled to the redistribution circuit structure, the redistribution circuit structure being between the first terminals and the insulating encapsulation;

second terminals, disposed on and electrically coupled to the conductive pillars, the solder resist layer being between the second terminals and the insulating encapsulation, wherein in a cross section of the package structure along a stacking direction of the insulating encapsulation and the solder resist layer, maximum lateral widths of the second terminals are less than maximum lateral widths of the plurality of openings, respectively, wherein in the cross section, a maximum lateral width of at least one of the plurality of openings is gradually increased along the stacking direction; and at least one passive device, connected to the redistribution circuit structure, wherein the first terminals and the at least one passive device are disposed at a side of the redistribution circuit structure.

2. The package structure of claim 1, wherein the plurality of openings penetrates through the solder resist layer from a first side of the solder resist layer to a second side of the solder resist layer, and the first side and the second side are opposite to each other,

wherein a first opening size of one or more than one of the plurality of openings at the first side is greater than a second opening size of one or more than one of the plurality of openings at the second side, wherein the first side is in contact with the insulating encapsulation, and the second side is free of the insulating encapsulation.

3. The package structure of claim 2, wherein a material of the solder resist layer is a photosensitive material.

4. The package structure of claim 1, wherein the plurality of openings penetrates through the solder resist layer from a first side of the solder resist layer to a second side of the solder resist layer, and the first side and the second side are opposite to each other,

wherein a first opening size of one or more than one of the plurality of openings at the first side is greater than a second opening size of one or more than one of the plurality of openings at the second side, wherein the second side is in contact with the insulating encapsulation, and the first side is free of the insulating encapsulation.

5. The package structure of claim 4, wherein a material of the solder resist layer is a non-photosensitive material.

6. The package structure of claim 1, wherein the surfaces of the conductive pillars are substantially coplanar to the second side of the solder resist layer.

7. The package structure of claim 1, further comprising: a sub-package, disposed on the solder resist layer and electrically coupled to the semiconductor die through the second terminals, the sub-package comprising at least one memory die encapsulated in an encapsulant.

8. The package structure of claim 1, further comprises a die attach film between the semiconductor die and the solder resist layer.

9. The package structure of claim 1, wherein in the stacking direction, a thickness of the second terminals is less than a thickness of the solder resist layer.

10. The package structure of claim 1, wherein in the cross section, sidewalls of at least one of the plurality of openings are slantly planar.

11. A package structure, comprising:
a semiconductor die;

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conductive pillars, next to and electrically coupled to the semiconductor die;

an insulating encapsulation, laterally encapsulating the semiconductor die and the conductive pillars, the insulating encapsulation having a first surface and a second surface opposite to the first surface;

a first redistribution circuit structure, disposed on the first surface of the insulating encapsulation and electrically coupled to the semiconductor die;

a second redistribution circuit structure, disposed on the second surface of the insulating encapsulation and electrically coupled to the semiconductor die, the conductive pillars electrically coupling the first redistribution circuit structure and the second redistribution circuit structure;

a solder resist layer, disposed on a side of the second redistribution circuit structure away from the insulating encapsulation, wherein the solder resist layer is a single-layer structure comprising a plurality of openings exposing a surface of a metallization layer of the second redistribution circuit structure, wherein each of the plurality of openings penetrates through the solder resist layer from a first side of the solder resist layer to a second side of the solder resist layer, and the first side and the second side are opposite to each other, wherein a first opening size of each of the plurality of openings at the first side is greater than a second opening size of each of the plurality of openings at the second side, and the surface of the metallization layer is substantially coplanar to the second side of the solder resist layer;

first terminals, disposed on and electrically coupled to the first redistribution circuit structure, the first redistribution circuit structure being between the first terminals and the insulating encapsulation; and

second terminals, disposed on the solder resist layer and electrically coupled to the conductive pillars, the solder resist layer being between the second terminals and the second redistribution circuit structure, wherein in a cross section of the package structure along a stacking direction of the insulating encapsulation and the solder resist layer, maximum lateral widths of the second terminals are less than maximum lateral widths of the plurality of openings, respectively.

12. The package structure of claim 11, wherein the first side is in contact with the second redistribution circuit structure, and the second side is free of the second redistribution circuit structure.

13. The package structure of claim 12, wherein a material of the solder resist layer is a photosensitive material.

14. The package structure of claim 11, wherein the second side is in contact with the second redistribution circuit structure, and the first side is free of the second redistribution circuit structure.

15. The package structure of claim 14, wherein a material of the solder resist layer is a non-photosensitive material.

16. The package structure of claim 11, further comprising at least one of:

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at least one passive device, connected to the first redistribution circuit structure, wherein the first terminals and the at least one passive device are disposed at a side of the first redistribution circuit structure; or

a die attach film, disposed between the semiconductor die and the solder resist layer.

17. The package structure of claim 11, further comprising:

a sub-package, disposed on the solder resist layer and electrically coupled to the semiconductor die through the second terminals, the sub-package comprising at least one memory die encapsulated in an encapsulant.

18. The package structure of claim 11, wherein in the stacking direction, a thickness of the second terminals is less than a thickness of the solder resist layer.

19. A package structure, comprising:

a semiconductor die;

conductive pillars, next to and electrically coupled to the semiconductor die;

an insulating encapsulation, laterally encapsulating the semiconductor die and the conductive pillars, the insulating encapsulation having a first surface and a second surface opposite to the first surface;

a first redistribution circuit structure, disposed over the first surface of the insulating encapsulation and electrically coupled to the semiconductor die;

a solder resist layer, disposed over the second surface of the insulating encapsulation, wherein the solder resist layer is a single-layer structure comprising a plurality of openings;

first terminals, disposed over and electrically coupled to the first redistribution circuit structure, the first redistribution circuit structure being between the first terminals and the insulating encapsulation; and

second terminals, disposed over the solder resist layer and electrically coupled to the conductive pillars, the solder resist layer being between the second terminals and the insulating encapsulation,

wherein in a cross section of the package structure along a stacking direction of the insulating encapsulation and the solder resist layer, maximum lateral widths of the second terminals are less than maximum lateral widths of the plurality of openings, respectively,

wherein in the cross section, sidewalls of at least one of the plurality of openings are slantly planar.

20. The package structure of claim 19, further comprising:

a second redistribution circuit structure, disposed between the insulating encapsulation and the solder resist layer and electrically coupled to the semiconductor die through the conductive pillars and the first redistribution circuit structure, the second redistribution circuit structure being electrically coupled to the second terminals; and

at least one passive device, connected to the first redistribution circuit structure, wherein the first terminals and the at least one passive device are disposed at a side of the first redistribution circuit structure.

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